

TPN Bridge Planning, Installation and Service

TN30-100a

R100

04/04

Notices and Trademarks

**Copyright 2000-2004 by Honeywell International Inc.
Release 100 April 2004**

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell and **TotalPlant** are registered trademarks of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell International
Process Solutions
2500 West Union Hills
Phoenix, AZ 85027
1-800 343-022

About This Document

Abstract

This publication supports the hardware planning, installation, and service of the TPN Bridge based upon the Dell OptiPlex GX200 and the WKS computer platforms.

References

Honeywell documents

The following list identifies all Honeywell documents that may be sources of reference for the material discussed in this publication.

Document Title
<i>Local Control Network Planning</i>
<i>TPN System Installation</i>
<i>TPS System Administration Guide for NT</i>
<i>Keyboards</i>
<i>Button Configuration Data Entry</i>
<i>TPN Bridge Overview and Implementation Guide</i>
<i>TPS System Implementation Guide</i>
<i>Planning, Installation and Service for TPN Connected Nodes (GX200)</i>
<i>Planning, Installation and Service for TPN Connected Nodes (WKS1/SVR1)</i>

Other documents

The following list identifies other documents that may be sources of reference for the material discussed in this publication.

Title
<i>Dell® OptiPlex® GX1 Systems Service manual</i>
<i>Dell® OptiPlex® GX1 Reference and Installation Guide</i>
<i>Dell Optiplex GXa Service Manual</i>
<i>Dell™ OptiPlex™ GX200 System User's Guide</i>
<i>Dell™ OptiPlex™ GX200 Service Manual</i>
<i>Dell® Diagnostic and Troubleshooting Guide - Dell 08220</i>
<i>ATXPII BIOS and Programmer's Reference Guide - MTAXBOSA/RM1</i>
<i>SyncMaster700B Owner's Manual - BH68-60707A</i>

Contacts

World Wide Web

The following Honeywell web sites may be of interest to Process Solution customers.

Honeywell Organization	WWW Address (URL)
Corporate	http://www.honeywell.com
Process Solutions	http://www.acs.honeywell.com
International	http://content.honeywell.com/global/

Other	WWW Address (URL)
Dell Computer Corporation	http://support.dell.com

Telephone

Contact us by telephone at the numbers listed below.

	Organization	Phone Number	
United States and Canada	Honeywell International Inc. Process Solutions	1-800-343-0228	Sales
		1-800-525-7439	Service
		1-800-822-7673	Technical Support
Asia Pacific	Honeywell Asia Pacific Inc. Hong Kong	(852) 23 31 9133	
Europe	Honeywell PACE Brussels, Belgium	[32-2] 728-2711	
Latin America	Honeywell International Inc. Sunrise, Florida U.S.A.	(954) 845-2600	

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation, which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

Symbol	Definition
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.
	Earth Ground: Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

Contents

1	PLANNING THE TPN BRIDGE PLATFORM.....	17
1.1	Network Architecture of TPN Bridge.....	17
	Scope.....	17
1.2	TPN Bridge System Time Link.....	18
1.3	Implementation of Three Segments with Time Link and GPS Option	19
	Scope.....	19
1.4	TPN Bridge Hardware	21
	Platform.....	21
	Description	21
	Ethernet switches.....	22
	Ethernet cables	22
	IP Addressing.....	22
	TPN Bridge model numbers.....	22
1.5	Hardware Model Numbers.....	23
	TPN Bridge Platforms	23
	Monitors	23
	Keyboards/mouse	23
	TPN Cabinet Mounting Hardware	23
	Shared User Interface Mounting	24
	Associated Models numbers	24
	GPS Options (Systems that Require Accurate Reference to UTC Time).....	25
1.6	Ethernet Switch	26
	Hardware.....	26
	Purchasing Information	26
1.7	Time Link Cable and GPS Options.....	27
	Scope	27
	Description	27
	Power/Surge Junction Box.....	27
	Antenna.....	28
	Antenna placement	29
	Cable, Antenna to Junction Box	30
1.8	Time Link Cables.....	31
	Scope.....	31
	Description	31
	Standard Cable Lengths	32
	Cable construction	32
	Other Time Link Components	33
	GPS Power Surge/Junction Box	34
	Signal and Connector Wiring.....	35

1.9	TPN Bridge LAN Cables	36
	Scope	36
	Description	36
	Standard Cable Lengths.....	37
	Cable construction.....	38
1.10	Equipment Configuration Rules	39
	Scope	39
	Model numbers.....	39
	Associated Model Numbers	42
1.11	TPN Bridge Configuration Rules	43
	Overview	43
	Mounting TPN Computers in LCN Cabinet.....	43
1.12	Mounting Ethernet Switch Modules in LCN Cabinet	44
	Overview	44
	Mouse	44
	Keyboard.....	44
	CRT Monitor.....	44
1.13	Human Interface Switch	45
	Shared Peripheral Switch.....	45
	Configuration rules	45
	Cabling	46
1.14	Allowable TPN Bridge Node Configurations	47
	Scope	47
1.15	Switch Module Positioning	48
	Scope	48
	TPN Cabinet mounting	48
	Port Connectors	49
1.16	Equipment Specifications	50
	Desktop (horizontal) GX200 Specification.....	50
	Deskside (vertical) GX1 Specification	50
	Desktop (horizontal) GX1 Specification.....	50
	Peripheral Interface Switch Specifications	51
1.17	TPN Bridge External Cabling	52
	Scope	52
1.18	Regulatory Compliance.....	53
	Desktop	53
	Safety Compliance	53
	Regulatory Compliance, Cabinet.....	54
	Safety Compliance	54
1.19	Ethernet Switch Specifications	55

Electrical specification	55
Physical dimensions	55
Environmental specifications	56
1.20 GPS Antenna System Specifications	57
1.21 Backup Power	59
2. INSTALLING THE TPN BRIDGE	61
2.1 Overview	61
Scope	61
Process description	61
2.2 Installing TPN Bridges	62
Scope	62
Locate Equipment	62
Grounding TPN Bridge Nodes	62
Connecting AC Power	63
2.3 TPN Bridge Cable Connections	64
GX200 Platform cable connections	64
WKS1 Platform cable connections	65
Dual NIC and LAN Cable connections	66
2.4 AC Power	67
TPN Bridge computer	67
TPN-Cabinet	67
2.5 Preparation for Installing the Ethernet Cable Network	69
Scope	69
2.6 Cable Connection Matrix	70
Scope	70
2.7 Installing the TPN Bridge	71
Procedure	71
2.8 Installing the TPN Bridge in a TPN Cabinet	72
Scope	72
2.9 Installing the GPS Antenna and/or Time Link Components	74
Scope	74
Install Time Link Cable Components	74
Install Power/Junction box (GPS Option)	75
Install GPS Antenna	76
2.10 Installing the Human Interface Switch	77
Scope	77

Connecting the User Console to the TPN Bridge	78
Selecting a Computer	79
3. SERVICE	81
3.1 Service Policy	81
Definition of support	81
Electronic chassis.....	81
3.2 Board Placement	82
Scope	82
Board Placement.....	82
3.3 Troubleshooting	83
Scope	83
Verifying Correct System Interrupt Assignments.....	83
3.4 Removing and Replacing the Computer in the Cabinet	84
Foam insulators.....	84
Remove the Computer from the Cabinet.....	84
Replace the Computer in the Cabinet	86
3.5 Fan Replacement	86
Procedure.....	86
4. TROUBLESHOOTING, CLEANING, AND PINNING DRIVES	87
4.1 Troubleshooting an Inoperative Drive.....	87
Failure modes.....	87
Isolation procedure.....	87
Illustration	87
4.2 Remove a Zip Cartridge with Power Off	88
Scope	88
4.3 Pinning the Drive SCSI Address	88
Scope	88
5. SERVICING PERIPHERAL DRIVES	89
5.1 Overview	89
Scope	89
5.2 Pinning the Drive SCSI Address	89
Drive SCSI address pinning	89
5.3 Replacing the Drive	90
Procedure.....	90

6.	DRIVE SERVICE	91
6.1	Overview	91
	Scope	91
	References	91
7.	17-INCH SAMSUNG MONITOR SERVICE	93
7.1	Overview	93
	Scope	93
7.2	Operational Controls	93
	Front Controls	93
	Front control panel	94
	Controls key table	95
7.3	Operational Controls Description.....	96
	Scope	96
	Power button	96
	Power indicator	96
	Contrast control.....	96
	Brightness control	96
	Adjustment control	96
8.	ETHERNET SERVICE.....	97
8.1	Troubleshooting.....	97
	Scope	97
	Illustration.....	97
9.	REPLACING AND SERVICING PARTS	99
9.1	Power Supply.....	99
	Overview	99
9.2	Dual NIC Adapter.....	99
	Replacing the Dual NIC adapter.....	99
	Updating the Dual NIC adapter driver	99
9.3	Spare Parts	101
	WKS1 Platform.....	101
	GX200 Platform.....	103
	GX1 Platform.....	104
	Ethernet Switch replacement unit.....	105

10. GLOSSARY	107
10.1 Communications Terminology	107
10.2 Computer System Terms.....	108
10.3 Honeywell Terminology.....	108
11. INDEX.....	109

Tables

Table 1-1	Antenna Cable Pin-Out.....	30
Table 1-2	Standard Cable Lengths for 100-Base-T Cable w/Black Boots	32
Table 1-3	Time Link T-Connector and Modular Terminator	33
Table 1-4	GPS Power Surge/Junction Box Signal Wiring Pinout.....	35
Table 1-5	Standard Cable Lengths for 100-Base-T Cable w/Yellow Boots	37
Table 1-6	Standard Cable Lengths for 100-Base-T Cable w/Green Boots	37
Table 1-7	TPN Bridge Model Numbers	39
Table 1-8	TPN Bridge Associated Model Numbers.....	42
Table 1-9	Desktop GX200 Specifications	50
Table 1-10	Desktop GX1 Specifications	50
Table 1-11	Desktop GX1 Specifications	50
Table 1-12	Electromagnetic Compatibility (EMC).....	53
Table 1-13	Electromagnetic Compatibility (EMC).....	54
Table 1-14	Nortel BayStack 450-12 Electrical Specifications	55
Table 1-15	Nortel BayStack 450-12 Physical Specifications.....	55
Table 1-16	Environment Specifications	56
Table 1-17	GPS Antenna Specification	57
Table 1-18	GPS System Power Supply Specification	59
Table 2-1	Cable Connection Matrix Template for TPN Bridge	70
Table 2-2	Connecting the Cables to the Computer	71
Table 2-3	Installation Parts (51403280) for the Computer Enclosure	72
Table 2-4	Installing the Computer Mounting Enclosure in a TPN Cabinet.....	73
Table 2-5	Installing the Time Link Cable Components.....	74
Table 2-6	Installing the Power/Surge Protect Junction Box	75
Table 2-7	Installing the GPS Antenna	76
Table 2-8	Installing the HI Switch(es).....	78
Table 2-9	Selecting a Computer by Switch Push-Button or Hot-Key	79
Table 2-10	Hot-Key Sequence for Selecting a Computer	79
Table 3-1	Board Placement.....	82
Table 3-2	Verifying System Interrupt Assignments	83
Table 3-3	System Interrupt Assignments	84

Table 3-4	Removing the Computer from the Cabinet	85
Table 3-5	Removing the Computer from the Cabinet	86
Table 4-1	Troubleshooting a Drive Problem	87
Table 4-2	SCSI Drive Addresses	88
Table 5-1	Replacing a Drive.....	90
Table 7-1	Key to Monitor 17-inch Controls.....	95
Table 9-1	Updating Dual NIC adapter driver.....	99
Table 9-2	PC Electronics Spare Parts for WKS1.....	101
Table 9-3	PC Electronics Spare Parts for GX200.....	103
Table 9-4	Computer Electronics Spare Parts GX1	104
Table 9-5	Unique Ethernet Switch Spare Parts	105

Figures

Figure 1-1	TPN Bridge Node in a TPN Segment	17
Figure 1-2	GPS and Time Link Connected to TPN Segments.....	18
Figure 1-3	Three Segments With GPS Option.....	20
Figure 1-4	Three Segments Configuration with Optional Timing References.....	21
Figure 1-5	Trimble Palisade Antenna Mounted on a Fiberglass Mast	28
Figure 1-6	Positioning the GPS Antenna	29
Figure 1-7	100 BaseTX Cable, 51305562-4XX.....	31
Figure 1-8	Cable Construction and Pin Assignments	32
Figure 1-9	GPS Power Surge/Junction Box Block Diagram	34
Figure 1-10	100 LAN BaseTX Cable, 51305482	36
Figure 1-11	Cable Construction and Pin Assignments.....	38
Figure 1-12	Cabling Shared Peripheral Interface Switch	46
Figure 1-13	ETHERNET Switch Mounting Module.....	48
Figure 1-14	Cabling for TPN Bridge Node.....	52
Figure 2-1	Ground Terminal in a TPN Cabinet	62
Figure 2-2	Cable Connections for GX200	64
Figure 2-3	Cable Connections for WKS1	66
Figure 2-4	Dual NIC Adapters for TPN Bridge	66
Figure 2-5	TPN Bridge Ethernet Cable Identification	69
Figure 3-1	Board Placement in TPN Bridge	82
Figure 4-1	Remote Peripheral Connector Pin-out.....	87
Figure 7-1	Samsung SyncMaster 17-Inch Monitor Front View	93
Figure 7-2	Front Panel Controls	94
Figure 8-1	Ethernet Indicators on the RJ-45 Connector	97

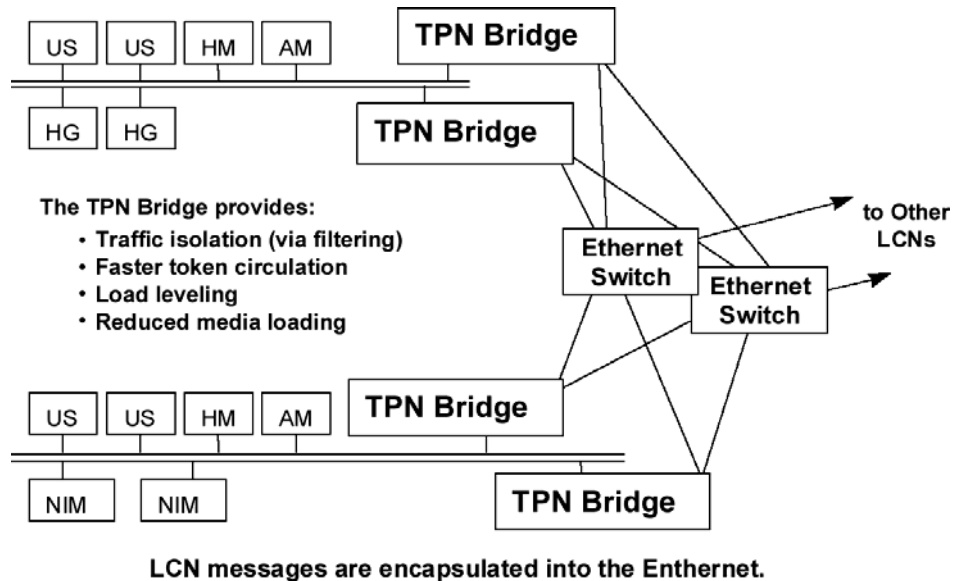
Contents

1 Planning the TPN Bridge Platform

1.1 Network Architecture of TPN Bridge

Scope

The TPN Bridge facilitates a smooth migration for existing large-system customers to TPS and into the future of Honeywell's networking technology. The TPN Bridge is not a LAN, but is a dedicated computer that reduces network traffic. The TPN Bridge is a device that transparently breaks up large LCN systems into segments compatible with this goal. The bridges provide improved LCN cluster load capacity because the bridge acts as a traffic filter, which eliminates unnecessary inter-segment traffic. This lowers the overall amount of traffic on a given LCN segment while speeding up the token circulation on each segment.



17861

Figure 1-1 TPN Bridge Node in a TPN Segment

1.2 TPN Bridge System Time Link

TPN time synchronization messages (Time Link) will be handled in a special manner providing rapid and more accurate time synchronization than can be achieved by treating the time messages as normal bridge traffic that simply passes through the bridges. The variable nature of the store and forward latency in the bridges will, however, not allow the current accuracy of TPN, single segment, Sequence of Events (SOE) timing.

The efficiency and accuracy of the system-wide time services can be increased using an optional time link cabling. Improving the long-term stability of system time is accomplished by means of a direct relationship to Universal Time Coordinated (UTC). This may be achieved using the optional addition of a Global Positioning System (GPS) antenna that is connected to redundant TPN Bridge nodes by means of the shielded twisted pair cable. This optional GPS system is shown in illustration below.

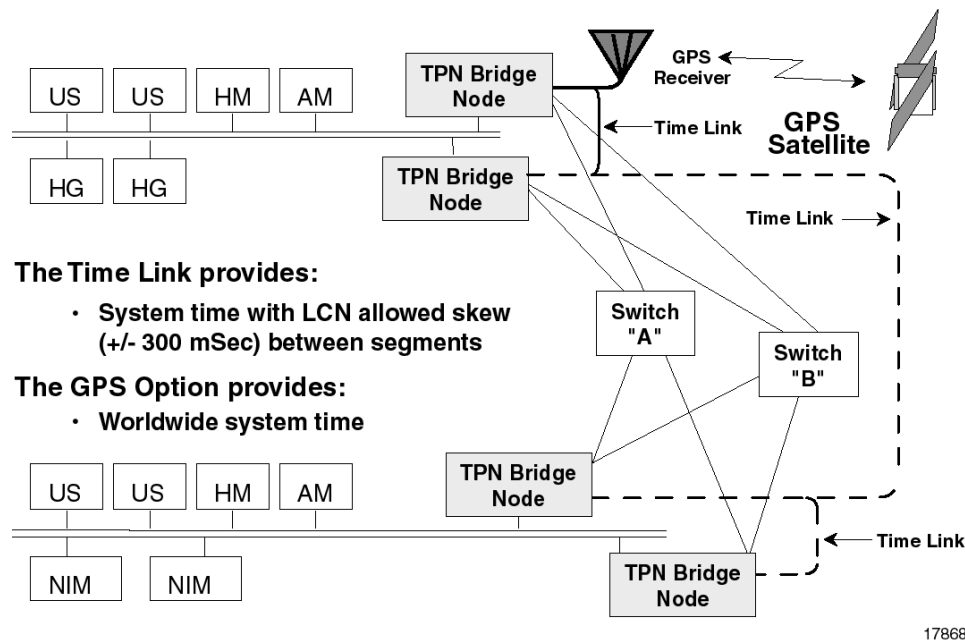


Figure 1-2 GPS and Time Link Connected to TPN Segments

1.3 Implementation of Three Segments with Time Link and GPS Option

Scope

A maximum of six segments of TPN nodes can be implemented using the TPN Bridge technology. An Ethernet switch module is dedicated to TPN Cable A and to TPN Cable B. Each TPN Bridge node is cabled to both switch modules as shown below. The Time Link and the GPS options are also shown.

1. Planning the TPN Bridge Platform

1.3. Implementation of Three Segments with Time Link and GPS Option

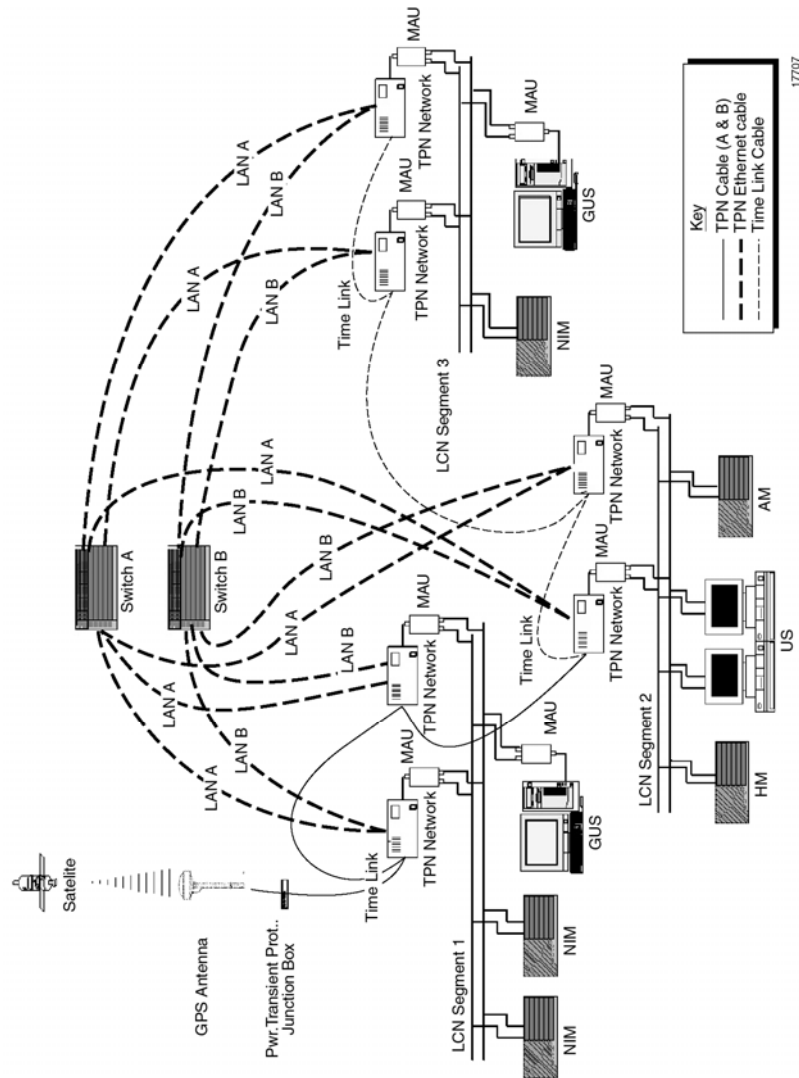


Figure 1-3 Three Segments With GPS Option

1.4 TPN Bridge Hardware

Platform

The hardware platform calls for redundant computers with a dual 10/100 BASE-T Ethernet interface, TPNB interface card and TPN Bridge software. TPN Bridge is intended for rack mounting in a cabinet to provide the highest reliability for the TPN Bridge operation.

Description

The TPN Bridge provides an increase in usable bandwidth for LCN clusters by providing a means of physically separating the LCN into segments and reducing the traffic on each segment to messages that involve elements of that segment. The pair of Ethernet Switches provides the redundant Bridge LAN network connections between each cluster.

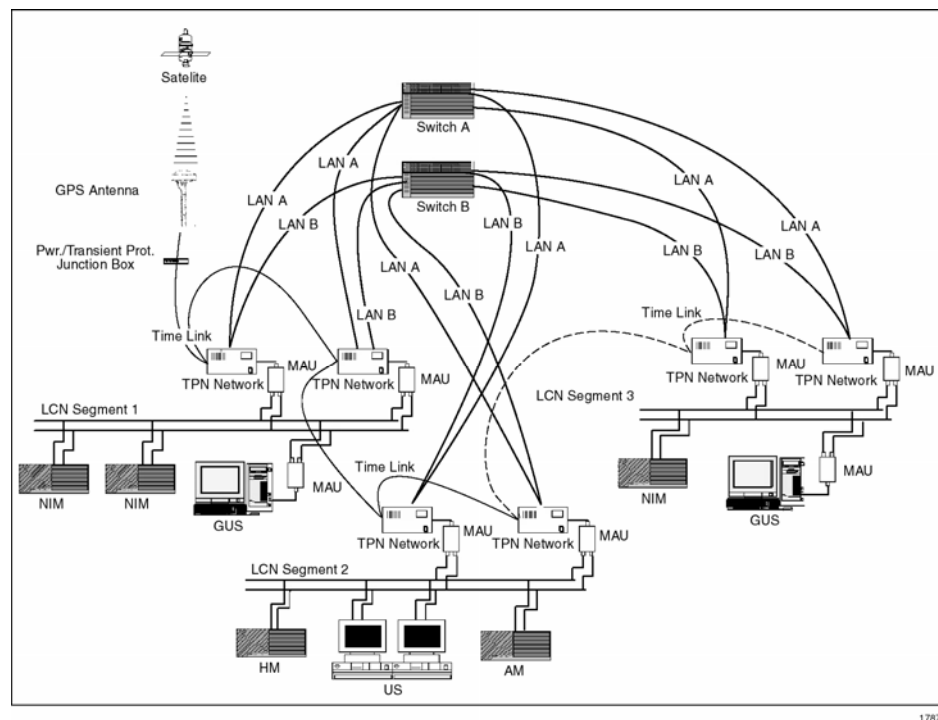


Figure 1-4 Three Segments Configuration with Optional Timing References

1. Planning the TPN Bridge Platform

1.4. TPN Bridge Hardware

Figure 1-4 shows a 3-segment configuration with both the optional Time Link cables for low cluster-to-cluster skew and the GPS timing reference. The user interface elements (shared monitors, keyboards, mice and peripheral switches) are not shown to improve clarity of the drawing from the perspective of the Bridge LAN and Time Link cable connections.

Ethernet switches

The Ethernet equipment for the TPN Bridge will consist of redundant 12 port switches. Two switches will be used to provide redundant paths for Cluster to Cluster communication. The Ethernet switches can be mounted in standard LCN cabinets.

Ethernet cables

The Ethernet cables will all be CAT-5 with an overall shield. The cables will be available in lengths from 1 - 99 meters. The cables will be terminated at both ends with shielded RJ-45 connectors. The RJ-45 connectors will have color coded boots on them at both ends of the cable to identify them as part of the TPN Bridge network.

- Yellow boots are used to identify cables within TPN Bridge LAN “A.”
- Green boots are used to identify cables within TPN Bridge LAN “B.”
- Black boots are used to identify cables within the TPN Bridge Time Link cable system.

IP Addressing

The TPN Bridge does not utilize a Primary Domain Controller (PDC) or Backup Domain Controller (BDC). The IP addresses for the TPN Bridge Dual NICs are fixed. The selection is from a reserved group of addresses and the individual assignments are performed when the Bridge nodes are configured. The IP selection is based upon a Bridge number identification system (BRIDGE1, BRIDGE2, BRIDGE3, etc.). Refer to the *TPN Bridge Overview and Implementation Guide* for details.

TPN Bridge model numbers

The model numbers for the TPN Bridge are structured so that the shared CRT monitor, Industrial CE-mark keyboard and Industrial CE-mark mouse are defined with separate model numbers.

1.5 Hardware Model Numbers

TPN Bridge Platforms

TP-BRGR01 2 WKS1 platforms, 2 TPNB PWAs, 2 LCN MAUs, 2 LCN MAU Cables, 2 Grounded Terminator, 2 CD Drive, 2 Dual NIC cards, TPN Bridge CD ROM

TP-TPNB04 2 Dell GX200 platforms, 2 TPNB PWAs, 2 LCN MAUs, 2 LCN MAU Cables, 2 Grounded Terminator, 2 CD Drive, 2 Dual NIC cards, TPN Bridge CD ROM

TP-TPNB01 2 Dell GX1 DeskSide platforms, 2 TPNB PWAs, 2 LCN MAUs, 2 LCN MAU Cables, 2 Grounded Terminator, 2 CD Drive, 2 Dual NIC cards, TPN Bridge CD ROM*

TP-TPNB02 2 Dell GX1 DeskTop platforms, 2 TPNB PWAs, 2 LCN MAUs, 2 LCN MAU Cables, 2 Grounded Terminator, 2 CD Drive, 2 Dual NIC cards, TPN Bridge CD ROM*

***Note:** these platforms are no longer being sold, but they are supported by Honeywell Dual Ethernet LAN Switch

TP-TPNBS1 2 Bay Network 12 port Ethernet Switches

Monitors

TP-DM17SA Desktop/Deskside 17" Monitor

Keyboards/mouse

TP-ICEKB1 QWERTY Keyboard, Industrial CE-Mark

TP-ICEMS1 Mouse, Industrial CE Mark

TPN Cabinet Mounting Hardware

TP-MOUNTR Kit for mounting a computer in a TPN Cabinet

TP-MRS12A Kit for mounting redundant (2) Ethernet Switches in TPN Cabinet

Tab 200 is for 120 Vac systems

Tab 201 is for 220 Vac systems

1. Planning the TPN Bridge Platform

1.5. Hardware Model Numbers

Shared User Interface Mounting

Multiple computers can use the same Monitor, Keyboard, and Mouse in a TPN Cabinet.

- | | |
|------------------|---|
| TP-RKVM4A | KVM SWITCH, 4-NODE, CABINET/RACK MOUNTING. This model calls out the Human Interface Switch, the appropriate cable set, and also provides the mechanical parts to mount the HI Switch, MONITOR, Keyboard and Mouse into the LCN cabinet. |
| TP-DKVM2A | KVM SWITCH, 2-NODE, DESKTOP. This model calls out the Human Interface Switch and the appropriate cable set for a two port desktop application. |
| TP-DKVM4A | KVM SWITCH, 4-NODE, DESKTOP. This model calls out the Human Interface Switch and the appropriate cable set for a four port desktop application. |

Associated Models numbers

- | | |
|---------------------|--|
| TP-HICAB1 | Parts for mounting the shared monitor, Keyboard, Mouse, and Peripheral Switch in a TPN cabinet |
| TP-DM17SA | Shared 17-inch monitor for Desktop/Deskside |
| TP- ICEKB1 | Shared Industrial CE-Mark QUERTY keyboard |
| TP-ICEMS1 | Shared Industrial CE-Mark mouse |
| 51197084-100 | 2-Port User Interface Peripheral Switch and interface cables |

GPS Options (Systems that Require Accurate Reference to UTC Time)

Note: Include from the Time Link Components list, Time Link cables, T-connectors and 1 of the 2 Terminator components with one of the following.

- | | |
|------------------|---|
| TP-GPS105 | Smart Antenna Option w/50 Ft Cable, Trimble Palisade 8 Ch. Smart Ant., Power/Surge Prot. J-Box, Power Cable, J-Box Mntg Brkt & Hdwe, Cable (Tab 100) - 5 TW Pair, 24AWG, Shielded CAT5 |
| TP-GPS110 | Smart Antenna Option w/100 Ft Cable, Trimble Palisade 8 Ch. Smart Ant., Power/Surge Prot. J-Box, Power Cable, J-Box Mntg Brkt & Hdwe, Cable (Tab 200) - 5 TW Pair, 24AWG, Shielded CAT5 |
| TP-GPS120 | Smart Antenna Option w/200 Ft Cable, Trimble Palisade 8 Ch. Smart Ant., Power/Surge Prot. J-Box, Power Cable, J-Box Mntg Brkt & Hdwe, Cable (Tab 300) - 5 TW Pair, 24AWG, Shielded CAT5 |

1. Planning the TPN Bridge Platform

1.6. Ethernet Switch

1.6 Ethernet Switch

Hardware

Honeywell Purchase Specification 51197083
Manufacturer Bay Networks (Nortel Networks)
Model # 450-12T
Description 12 port Ethernet switch

Purchasing Information

TP-TPNBS1 12 Port 10/100BaseT Ethernet Switch (51197083-100)

Attention

All connections to the ETHERNET switch must be direct connections. **No punch-down closets are allowed.**

1.7 Time Link Cable and GPS Options

Scope

A Time Link cable option may be utilized to provide low skew between LCN segments.
A GPS antenna option may be utilized to provide accurate system time relative to UTC.

Four possibilities in Time Synchronization Accuracy in Bridge Application are:

- Option 1: No cable connections in terms of time link cabling. There is no GPS connection. This option only uses the software for segment synchronization. This is the simplest configuration and the least accurate (+/- 16 ms max skew between segments – No UTC reference for system).
- Option 2: Using a time link cable between the segments that provides low skew between segments. There is no reference to a GPS or Universal Time Clock. A low skew case but no reference to universal time. (+/- 300 uS max skew between segments – No UTC reference for system).
- Option 3: Use of GPS connected to bridges on a single segment acting as a reference to Universal Time Clock but still distributing the time to other segments using the software. This provides good system time accuracy but also a fairly high segment-to-segment skew (+/- 16 mS max skew between segments – System is referenced to UTC)
- Option 4: Use of both GPS connection and time link cable connection to all segments. This provides low skew segment to segment and an accurate system time relative to UTC. (+/- 300 uS max skew between segments – System is referenced to UTC.)

Description

The Time Link distribution cable consists of CAT5 STP cables, Tee connectors and terminators. The GPS antenna option consists of the GPS antenna, power surge junction box, and cable between the antenna and the junction box.

Power/Surge Junction Box

Does the following:

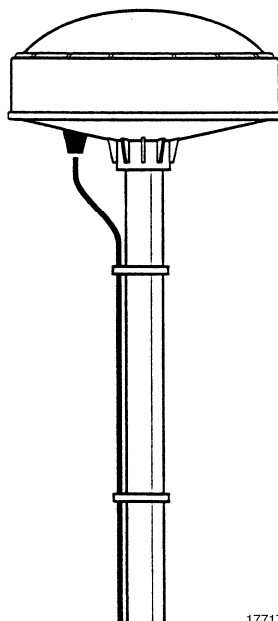
- provides DC power to the antenna receiver
- isolates the TPN system from power surges that may be induced in the antenna .

1. Planning the TPN Bridge Platform

1.7. Time Link Cable and GPS Options

Antenna

The antenna, shown below, is a low profile, easy to mount device that must have line-of-site positioning to the satellites. The junction box can be located up to 61 meters (200 feet) from the antenna.



17717

Figure 1-5 Trimble Palisade Antenna Mounted on a Fiberglass Mast

Antenna placement

The antenna must be positioned with uninterrupted line of sight of the Satellite. This means that the area 30° above the horizon must be 100% clear.

Since the antenna itself contains active electronic components that could be damaged by direct lightning strikes, the antenna should be positioned with the intent of making it an unattractive path for lightning discharge. This may include the use of a suitable lightning rod discharge element with suitable ground wire. If a lightning rod is utilized, the GPS antenna should be positioned within the protective cone where the radius (r) of the base of the cone is equal to the height (h) of the top of the lightning rod.

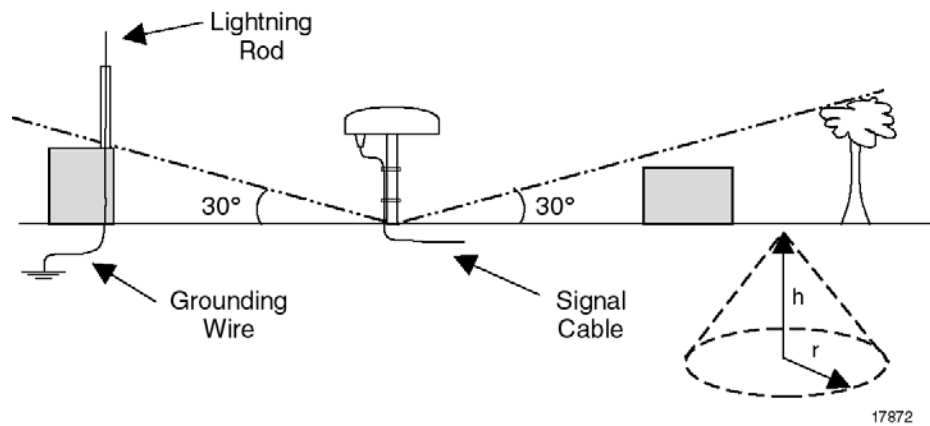


Figure 1-6 Positioning the GPS Antenna

1. Planning the TPN Bridge Platform

1.7. Time Link Cable and GPS Options

Cable, Antenna to Junction Box

This cable, 51305554, is constructed with six twisted pair conductors, shielded and covered with UV stabilizer/inhibitor insulation. The cable comes in lengths of 15.2 meters (50 feet), 30.5 meters (100 feet) or 61 meters (200 feet).

Table 1-1 Antenna Cable Pin-Out

Wire Type	Signal RS422	Deutsch Pin #	Color	DB9 pin #
Paired	Power	1	RED	9
	Ground	9	BLACK	5 (note 1)
	———	—	BLACK	1 (note 1)
Paired	Port B Rcv -	2	WHITE	8
	Port B Rcv +	3	BLACK	11
Paired	Port B Tx -	4	YELLOW	3
	Port B Tx +	5	BLACK	10
Paired	Port A Rcv -	6	- - -	NC
	Port A Rcv +	7	- - -	NC
Paired	Port A Tx -	8	GREEN	2
	Port A Tx +	10	BLACK	7
Paired	1PPS +	11	BLUE	4
	1PPS -	12	BLACK	6
Drain Wires	NC	NC	Un-insulated	Shield

Note 1: This is a loopback wire for DB09 connector only.

1.8 Time Link Cables

Scope

There are two sets of redundant TPN Bridge LAN Ethernet cable, each identified by a colored plastic ring (boot) around each end of the cable. The 100-Base-T cables (CAT5 w/RJ45 Shielded Plugs) Time Link cable is identified with a color code of Black.

Description

Time Link cables are for systems that require low segment-to-segment clock skew. The cable is uniquely identified by a ferrite bead inside adhesive shrink tubing at both ends.

The Time Link cable is available two ways: in standard lengths of 0.5, 2, 5, 10, 20, 50, 75, and 99 meters or custom specified lengths. All cables will have color-code of Black for identification boots (Tab 4XX where XX equals cable length) around each cable-end.

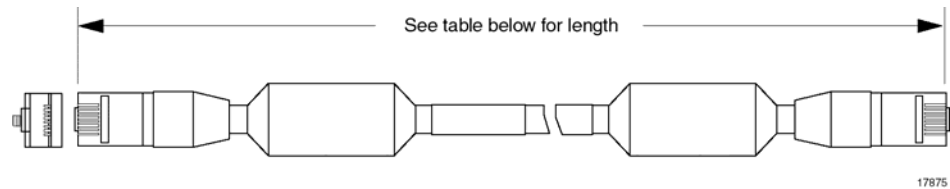


Figure 1-7 100 BaseTX Cable, 51305562-4XX

1. Planning the TPN Bridge Platform

1.8. Time Link Cables

Standard Cable Lengths

Standard length cables are available by factory order.

Table 1-2 Standard Cable Lengths for 100-Base-T Cable w/Black Boots

51305482-4XX Cable Tab	Standard Lengths
400	0.5 meters
402	2 meters
405	5 meters
410	10 meters
420	20 meters
450	50 meters
475	75 meters
499	99 meters
4XX	Custom Length XX meters

Cable construction

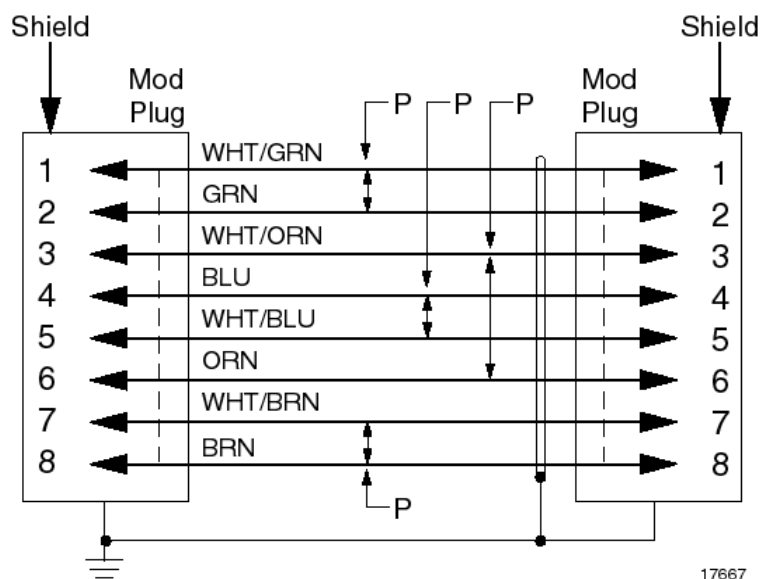


Figure 1-8 Cable Construction and Pin Assignments

Other Time Link Components

Table 1-3 Time Link T-Connector and Modular Terminator

QTY	Description	Part Number
2	RJ45 (8x8) Modular Plug Terminator (100 Ohm)	51192707-100
AR	T Connector - Receptacle RJ45 - 2x8P8C to 8P8C Shielded	51192708-100

1. Planning the TPN Bridge Platform

1.8. Time Link Cables

GPS Power Surge/Junction Box

The GPS Power Surge/Junction Box provides dc power to the GPS Antenna, ground connection for the antenna, and drives the Time Link signal distribution.

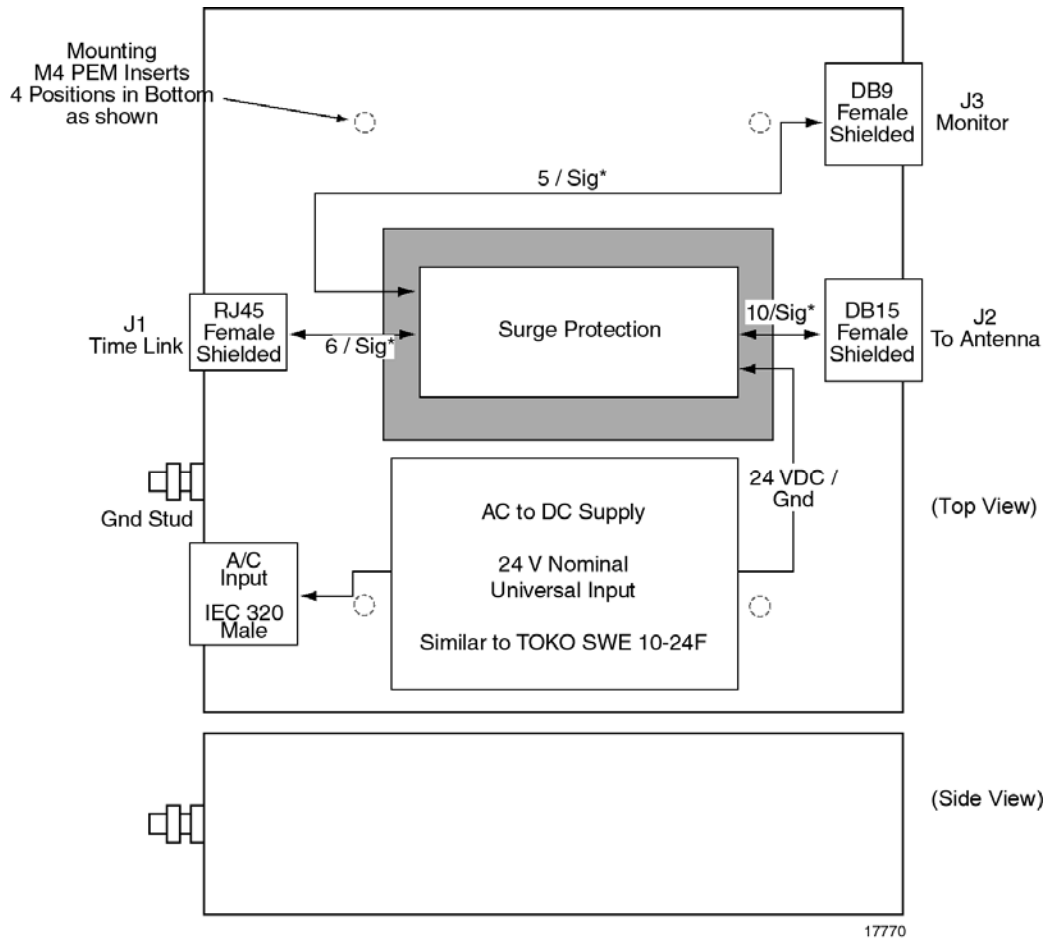


Figure 1-9 GPS Power Surge/Junction Box Block Diagram

- Notes:**
1. Signals are RS422 levels, low frequency (9600 Baud)
 2. DB15 connector end goes to GPS antenna, RJ45 end goes to the computer
 3. DB9 connector end is a monitor port used by service
 4. A mounting Bracket is required for wall mounting

Signal and Connector Wiring

The detailed pinout of each cable inter-connection is found in the following table.

Table 1-4 GPS Power Surge/Junction Box Signal Wiring Pinout

RJ45 J1 Pin #	Signal Description	DB15 J2 Pin #	Signal Description	DB9 J3 Pin #
1	N.C. (No Connection)	3	Port B Tx -	2
- - -	N.C. (No Connection)	10	Port B Tx+	6
2	GPS_DETECT	1	N.C. (No Connection)	1
3	1PPS+	4	N.C. (No Connection)	4
4	TSIP-	2	N.C. (No Connection)	8
5	TSIP+	7	N.C. (No Connection)	- - -
6	1PPS-	6	N.C. (No Connection)	- - -
7	N.C. (No Connection)	8	Port B Rcv -	3
- - -	N.C. (No Connection)	11	Port B Rcv +	7
8	Signal Ground / Power Common	5	Signal Ground Common	5
- - -	Power (+24 VDC)	9	N.C. (No Connection)	9
- - -	N.C. (No Connection)	12	N.C. (No Connection)	- - -
- - -	N.C. (No Connection)	13	N.C. (No Connection)	- - -
- - -	N.C. (No Connection)	14	N.C. (No Connection)	- - -
- - -	N.C. (No Connection)	15	N.C. (No Connection)	- - -
Shield	Signal Ground / Case (Pwr common)	Shield		Shield

1. Planning the TPN Bridge Platform

1.9. TPN Bridge LAN Cables

1.9 TPN Bridge LAN Cables

Scope

There are two sets of redundant TPN Bridge LAN Ethernet cables, each identified by a colored plastic ring (boot) around each end of the cable.

- Yellow boot identifies the Ethernet LAN Cable A (part number 51305482-1xx, where xx indicates length)
- Green boot identifies the Ethernet LAN Cable B (part number 51305482-2xx, where xx indicates length)

Standard lengths of cable are available in lengths of 2, 5, 10, 20, 50, 75 and 99 meters.

Description

These cables are available two ways: in standard lengths of 2, 5, 10, 20, 50, 75 and 99 meters or custom specified lengths. All cables have color-coded identification boots around each cable-end Ethernet connector.

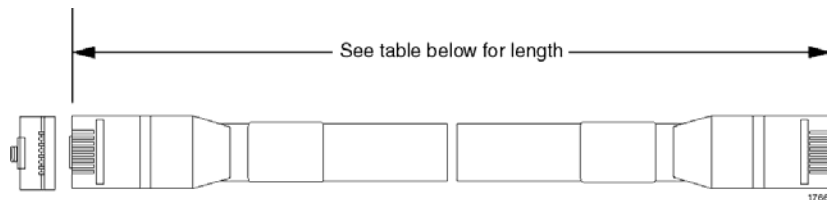


Figure 1-10 100 LAN BaseTX Cable, 51305482

Standard Cable Lengths

Standard length cables are available by factory order.

**Table 1-5 Standard Cable Lengths for 100-Base-T Cable
w/Yellow Boots**

51305482-1XX Cable Tab	Standard Lengths
102	2 meters
105	5 meters
110	10 meters
120	20 meters
150	50 meters
175	75 meters
199	99 meters
1XX	Custom Length XX meters

**Table 1-6 Standard Cable Lengths for 100-Base-T Cable
w/Green Boots**

51305482-2XX Cable Tab	Standard Lengths
202	2 meters
205	5 meters
220	10 meters
220	20 meters
250	50 meters
275	75 meters
299	99 meters
2XX	Custom Length XX meters

1. Planning the TPN Bridge Platform

1.9. TPN Bridge LAN Cables

Cable construction

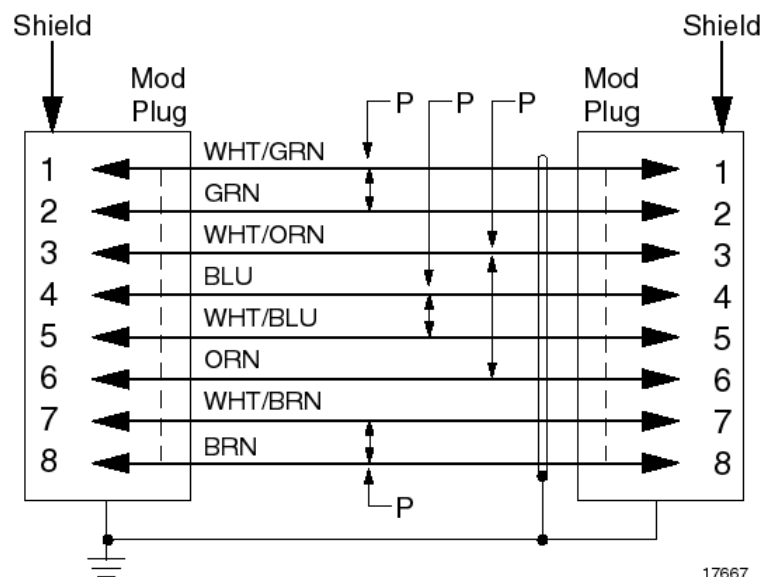


Figure 1-11 Cable Construction and Pin Assignments

1.10 Equipment Configuration Rules

Scope

The configuration rules control the placement of nodes, monitors, keyboards, switch modules, etc., so that there aren't any conflicts.

Model numbers

Table 1-7 TPN Bridge Model Numbers

Model Number	Description
TP- BRGR01	Redundant TPN Bridge. This model number calls for redundant (2) WKS1 computers each with a TPNB pwa, the LCN MAU and MAU cable, a pair of grounded terminators, a Dual NIC card, and a CD which contains the TPN Bridge personality, traffic monitor and analysis tool. With each of these model numbers the appropriate (2) computer Mounting Kits must be ordered (LCN Cabinet, Classic, Z or EZ furniture). The intended mounting scenario for the Bridge computers determines if a CRT, Keyboard, and Mouse must be ordered. If the user interface is to be mounted in an LCN cabinet, then the TP-RKVM4A (Shared User I/F Mounting Kit for LCN Cabinet) provides the mounting hardware for the user interface equipment.
TP-TPNB04	Redundant TPN Bridge. This model number calls for redundant (2) DELL GX200 computers each with a TPNB pwa, the LCN MAU and MAU cable, a pair of grounded terminators, a Dual NIC card, and a CD which contains the TPN Bridge personality, traffic monitor and analysis tool. With each of these model numbers the appropriate (2) computer Mounting Kits must be ordered (LCN Cabinet, Classic, Z or EZ furniture). The intended mounting scenario for the Bridge computers determines if a CRT, Keyboard, and Mouse must be ordered. If the user interface is to be mounted in an LCN cabinet, then the TP-RKVM4A (Shared User I/F Mounting Kit for LCN Cabinet) provides the mounting hardware for the user interface equipment.

1. Planning the TPN Bridge Platform

1.10. Equipment Configuration Rules

Table 1-7 TPN Bridge Model Numbers

Model Number	Description
TP-TPNB02	Redundant TPN Bridge. This model number calls for redundant (2) DELL Gx1 Bridge Desktop computers each with a TPNB pwa, the LCN MAU and MAU cable, a pair of grounded terminators, a Dual NIC card, and a CD which contains the TPN Bridge personality, traffic monitor and analysis tool. With each of these model numbers the appropriate (2) computer Mounting Kits must be ordered (LCN Cabinet, Classic, Z or EZ furniture). The intended mounting scenario for the Bridge computers determines if a CRT, Keyboard, and Mouse must be ordered. If the user interface is to be mounted in an LCN cabinet, then the TP-RKVM4A (Shared User I/F Mounting Kit for LCN Cabinet) provides the mounting hardware for the user interface equipment.
TP-TPNB01	Redundant TPN Bridge. This model number calls for redundant (2) DELL Gx1 Bridge Deskside computers each with a TPNB pwa, the LCN MAU and MAU cable, a pair of grounded terminators, a Dual NIC card, and a CD which contains the TPN Bridge personality, traffic monitor and analysis tool. These computers are intended to be utilized in applications other than LCN cabinet, Classic, Z or EZ furniture. In these applications the CRT(s), Keyboard(s), and Mouse(Mice) are ordered separately to provide the user interface for the redundant pair of Bridge nodes. The Keyboard/Video/Mouse (KVM) may be shared through a peripheral switch (TP-DKVM2A) or the user may order a Keyboard, Video and Mouse for each computer if desired.
TP-RKVM4A	Shared User I/F Mounting Kit for LCN Cabinet. Provides the mounting hardware for the user interface equipment.
TP-MOUNTR	Bridge Desktop computer Mounting Kit for LCN Cabinet. This model is used for mounting each Desktop TPN Bridge computer into LCN cabinets (rack mount).
TP-MRS12A	TPN Bridge Switch Mounting for TPN Cabinet. This model defines the pieces necessary to mount the Ethernet switch modules in an TPN cabinet (rack mount). Tab 100 mounts one switch in 120 Vac systems Tab 101 mounts one switch in 220 Vac systems Tab 200 mounts two switches in 120 Vac system Tab 201 mounts two switches in 220 Vac system

Table 1-7 TPN Bridge Model Numbers

Model Number	Description
TP-PCNXMA	TPN Bridge Expansion kit to add a second switch, contain the second switch, two expansion modules, and two cascade cables. Tab 100 is for 120 Vac versions Tab 200 is for 220 Vac versions
51153372	TPN BRIDGE Software Bundle. This part number for the CD-ROM that contains the Software utilized in TPN BRIDGE.

1. Planning the TPN Bridge Platform

1.10. Equipment Configuration Rules

Associated Model Numbers

The following are model numbers associated with the TPN BRIDGE.

Table 1-8 TPN Bridge Associated Model Numbers	
Model Number	Description
TP-DM17SA	Desktop / Deskside 17" MONITOR
TP-ICEKB1	QWERTY Keyboard Industrial CE-mark
TP-ICEMS1	Mouse Industrial CE-mark

1.11 TPN Bridge Configuration Rules

Overview

The TPN Bridge computer is intended to be installed in an LCN cabinet to provide the highest reliability of TNP Bridge operation.

The configuration constraints on the TPN Bridge computer (as well as other elements of the TPN Bridge system) are primarily related to prevention of the loss of cluster to cluster communication under conditions such as a tripped A/C circuit breaker or other single point of failure.

The following general rules apply to the allowed placement of the TPN Bridge computer, the redundant Ethernet Switches and the Peripheral Switch placement and connectivity.

1. Primary and secondary TPN Bridge node (Partners) and the redundant Ethernet Switches must be on separate AC power sources.
2. For Cabinet mounted equipment, the peripheral Switch, switched devices and the node computers must all be in the same cabinet. The human interface (Keyboard/Video/Mouse) cables can not go between cabinets.
3. The four port Peripheral Switch is used in cabinets.
4. If possible, multiple primary TPN Bridges should share a set of peripherals and the secondary bridges should share a separate set of peripherals. This may not be possible due to equipment layout. It is acceptable for a single bridge pair (primary and secondary) to share the same set of peripherals.

Mounting TPN Computers in LCN Cabinet

The TPN Bridge computer is mounted in an LCN cabinet (rack mounted). Since dual AC power entries are provided in the LCN cabinet, both the Primary and Secondary redundant Bridge partners may be mounted in the same LCN cabinet (as well as both Ethernet Switches). Another method is to position multiple Primary Bridge nodes in one LCN cabinet and multiple Secondary Bridge nodes in a second LCN cabinet. The TP-MOUNTR model provides the mechanical parts to mount each TPN Bridge computer in the cabinet.

1. Planning the TPN Bridge Platform

1.12. Mounting Ethernet Switch Modules in LCN Cabinet

1.12 Mounting Ethernet Switch Modules in LCN Cabinet

Overview

The Ethernet switches are installed in LCN Cabinets or on a desktop. Redundant Ethernet switches must not be connected to the same AC power source.

Mouse

The TPN Bridge computers shall utilize the industrial CE mouse (TP-ICEKMS1) currently specified for industrial TPS applications. One mouse will be used to operate multiple Bridge nodes through a Peripheral Switch.

Keyboard

The TPN Bridge computers shall utilize the industrial CE keyboard (TP-ICEKB1) currently specified for industrial TPS applications. One keyboard will be used to operate multiple Bridge nodes through a Peripheral Switch.

CRT Monitor

The 17-inch desktop CRT shall be used in all applications of the TPN Bridge. Since all TPN Bridges are mounted in an LCN cabinet, the CRT will be placed on a CRT shelf that holds the CRT, the Keyboard, the Mouse and the Peripheral Switch. Model TP-HICAB1 calls for the hardware to mount the shared CRT, Keyboard, Mouse and a 4 port Peripheral Switch. Note that no touch screen option is available for the 17-inch CRT.

1.13 Human Interface Switch

Shared Peripheral Switch

The Peripheral switch provides the method of allowing a user to interface with redundant TPN Bridge computer partners using a single CRT, Keyboard and Mouse. If a user wishes to utilize shared keyboard/video/mouse in the LCN cabinet configuration, model number TP-RKVM4A is required. All of the mounting hardware for the CRT, Keyboard, Mouse and a 4 port peripheral switch are included in this TP-RKVM4A Model number. The cables (Keyboard, Mouse, CRT cables) are approximately 6 feet in length and therefore are only long enough to connect from the TPN Bridge nodes to the peripheral switch within the same cabinet. This also applies to the cables that connect the peripheral switch to the Keyboard, Mouse and CRT.

Configuration rules

- The Human Interface Switch, the switched peripherals and the node computers must all be in the same cabinet. This is because the keyboard video and mouse cables are only 6 feet long.
- Cabinets support the four port switch only. In cases where only two ports will be implemented, the four port switch will be used without connections to the third and fourth ports.
- Concerning TPN Bridge – If possible, multiple primary TPN Bridge nodes should share a set of peripherals and the secondary bridges should share a separate set of peripherals. This may not be possible due to equipment layout. It is acceptable for a single bridge pair (primary and secondary) to share the same set of peripherals.

1. Planning the TPN Bridge Platform

1.13. Human Interface Switch

Cabling

The cables that connect the Human Interface Switch to the computers are only long enough to reach computers installed in the same cabinet as the HI Switch.

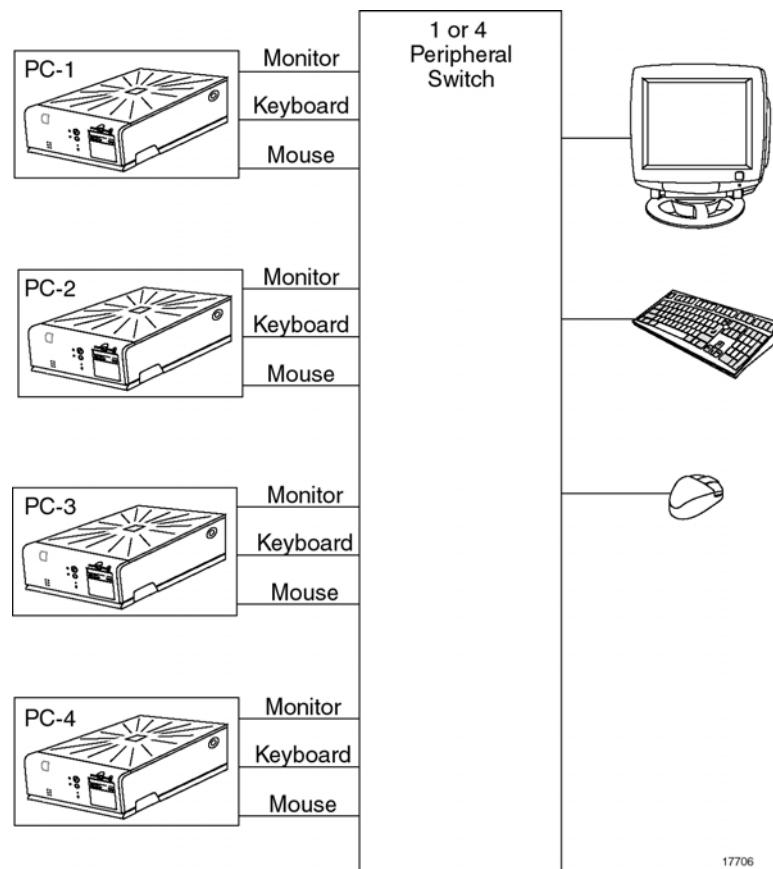


Figure 1-12 Cabling Shared Peripheral Interface Switch

1.14 Allowable TPN Bridge Node Configurations

Scope

The following configuration rules must be followed. When orders are placed in the Configurator, these rules are enforced.

The TPN Bridge node can be located:

Desktop (horizontal)	On a table top in the standard computer arrangement
Deskside (tower)	On the floor or on the worksurface top
TPN Cabinet	A mounting module located in a TPN cabinet. The 17-Inch monitor is mounted on top of the module. The keyboard tray flips down and has a mouse pad that is located to the right or left of the keyboard.

1.15 Switch Module Positioning

Scope

One or two switch modules can be located in a standard 19-inch rack mount.

Desktop	On a table top in a stacked arrangement
TPN Cabinet	A standard 19-Inch rack mount

TPN Cabinet mounting

The ETHERNET switch is mounted in the upper module position in a NEMA 19-Inch mounting module that can hold two switch units.

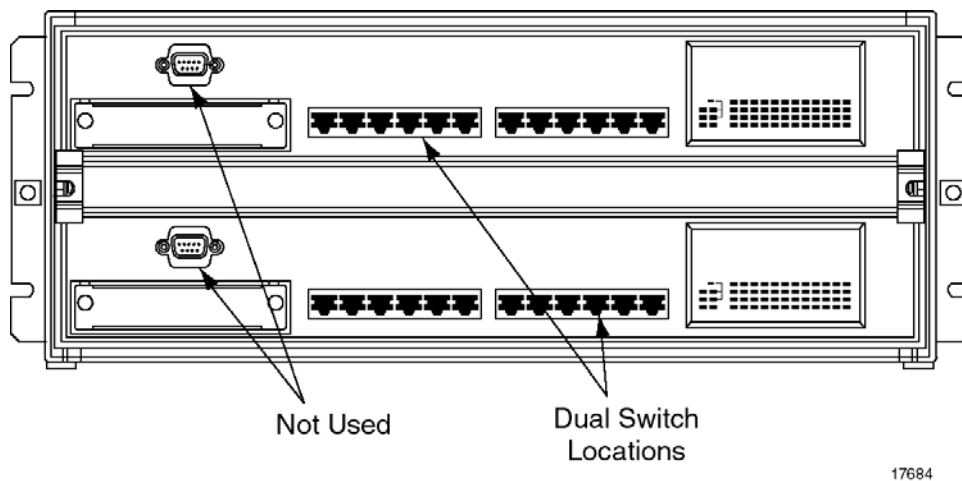


Figure 1-13 ETHERNET Switch Mounting Module

Port Connectors

The Ethernet switch uses 10BASE-T/100BASE-TX RJ-45 (8-pin modular) port connectors. There are 24 port connectors on the Ethernet switch.

The Ethernet switch is shipped with port connectors configured as MDI-X (media-dependent interface-crossover). These ports connect over straight cables to the dual network interface controller (NIC) card in the TPN Bridge computer.

1.16 Equipment Specifications

Desktop (horizontal) GX200 Specification

Dell OptiPlex GX200 specification can be found on the Dell Web site. You can view the specification and other documentation at <http://support.dell.com/us/en/docs>. You can also view Dell online documentation in the **Dell Accessories** folder on your computer. You may have received some hard copies of the documentation with the shipment.

Table 1-9 Desktop GX200 Specifications

Height	20.57 cm (8.1 inches)
Width	44.45 cm (17.5 inches)
Depth	43.68 cm (17.2 inches)
Weight	15.87 Kg (35 pounds)

Deskside (vertical) GX1 Specification

Refer to the Dell® OptiPlex™ GX1 specification LCA(C9860)897. To order a copy, use order number L1070-8300 for all specifications.

Table 1-10 Deskside GX1 Specifications

Height	44.45 cm (17.5 inches)
Width	20.57 cm (8.1 inches)
Depth	43.68 cm (17.2 inches)
Weight	15.87 Kg (35 pounds)

Desktop (horizontal) GX1 Specification

Table 1-11 Desktop GX1 Specifications

Height	16.5 cm (6.5 inches)
Width	42.09 cm (16.5 inches)
Depth	44.45 cm (17.5 inches)
Weight	15.87 Kg (35 pounds)

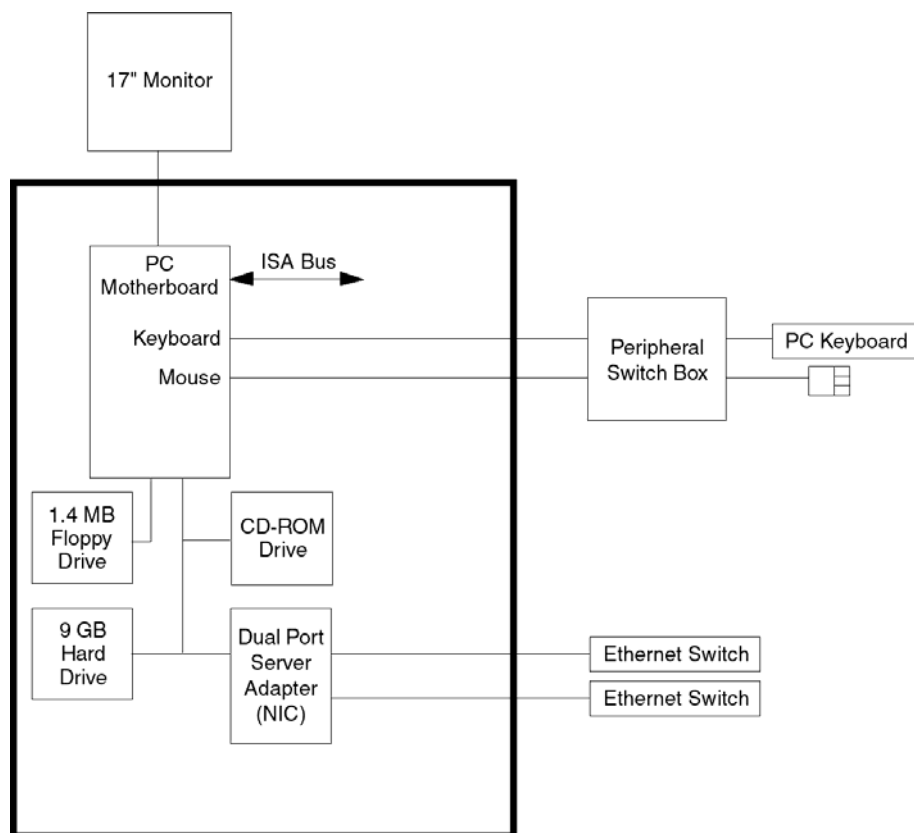
Peripheral Interface Switch Specifications

SV 2 Dimensions	Height: 4.83 cm (1.9 Inches) Width: 20.57 cm (8.1 Inches) Depth: 12.7 cm (5.0 Inches)
SV 4 Dimensions	Height: 6.86 cm (2.7 Inches) Width: 20.57 cm (8.1 Inches) Depth: 12.7 cm (5.0 Inches)
Environment/ Optional Power	Operating Temperature: 5° C to 41° C (41° to 104°) Storage Temperature: -20° C to 50° C (-4° to 122°) Optional Power 6 Vdc at 700 mA

1.17 TPN Bridge External Cabling

Scope

The TPN Bridge Node is equipped with, but can share, a 17-Inch monitor, 104 key Keyboard and a mouse



17705

Figure 1-14 Cabling for TPN Bridge Node

1.18 Regulatory Compliance

Desktop

Table 1-12 Electromagnetic Compatibility (EMC)

European Community:

Emissions: EN50081-1, 1992 (EN55022 Class A)

Immunity: EN50082-1, 1992

United States, Federal Communications Commission:

Emissions: FCC Part 15, Class A, 1996

Attention: The following formula is a proximity guideline, for use of Portable Transceivers

(walkie-talkies), in the frequency range of 80MHz to 1GHz:

D > **SQRT{P}** (D must be greater than the square root of P)

D = Distance from equipment, in meters.

P = Power Output of the Portable Transceivers (walkie-talkies), in Watts.

Examples:

P = 10 Watts, D > 3.162 meters

P = 5 Watts, D > 2.236 meters

P = 1 Watt, D > 1.000 meters

Note: Electrical cables, which are routed external to the equipment, must be fully shielded cables (360 degree metallic shielding) in order to comply with the above EMC standards.

Safety Compliance

Product Safety Compliance

CAN/CSA-C22.2 No. 950

EN60950

Note: Within these standards (above) is a 'Normative Reference' section citing additional standards which may apply as suited and required for product compliance.

Regulatory Compliance, Cabinet

Table 1-13 Electromagnetic Compatibility (EMC)

European Community:

Emissions: EN50081-2, 1994

Immunity: EN50082-2, 1995

Australian Communications Authority:

Emissions: AS/NZS 2064-1997 (CISPR 11, Class A) (SCN N314)

Attention: The following formula is a proximity guideline, for use of Portable Transceivers (walkie-talkies), in the frequency range of 80MHz to 1GHz:

$D > 0.30 \cdot \sqrt{P}$ (D must be greater than 0.30 multiplied by the square root of P)

D = Distance from equipment, in meters.

P = Power Output of the Portable Transceivers (walkie-talkies), in Watts.

Examples:

P = 10 Watts, $D > 0.949$ meters

P = 5 Watts, $D > 0.671$ meters

P = 1 Watt, $D > 0.300$ meter

Note: Electrical cables, which are routed external to the equipment, must be fully shielded cables (360 degree metallic shielding) in order to comply with the above EMC standards.

Safety Compliance

Product Safety Compliance

CAN/CSA-C22.2 No. 1010.1

EN61010-1

Note: Within these standards (above) is a 'Normative Reference' section citing additional standards which may apply as suited and required for product compliance.

WARNING

Honeywell IAC does not claim Safety Compliance or Electromagnetic Compatibility (EMC) Compliance for system equipment configurations that have not been described in this manual as standard system configurations. Any equipment configuration other than that described in this publication decertifies the Safety and EMC compliance of this product.

1.19 Ethernet Switch Specifications

Electrical specification

Table 1-14 Nortel BayStack 450-12 Electrical Specifications

Line Frequency	47 to 65 Hz
Input Volt Amperes Rating	120 W maximum
Input Voltage	100 – 240 Vac
Input Current	1.5 A @ 100 Vac
Thermal Rating	410 Btu/hr

Physical dimensions

Table 1-15 Nortel BayStack 450-12 Physical Specifications

Height	7.03cm (2.77 inches)
Width	44.07 cm (17.55 inches)
Depth	38.1 cm (15.0 inches)
Weight	5.26 KG (11.6 lb.)

1. Planning the TPN Bridge Platform

1.19. Ethernet Switch Specifications

Environmental specifications

Table 1-16 Environment Specifications

Operating Temperature	5°to 40°C (41°to 104° F)
Storage Temperature	-25° to 70° C (-13° to158° F)
Operating Humidity	85% maximum relative humidity, noncondensing
Storage Humidity	95% maximum relative humidity, noncondensing
Operating Altitude	3000 m (10,000 ft) maximum
Storage Altitude	3000 m (10,000 ft) maximum
Vibration	IEC 68-2-6/34
Shock/Bump	IEC 68-2-27-29
Electromagnetic Emissions	CE Compliant
Electromagnetic Susceptibility	
Electrostatic Discharge (ESD)	EC 801-2, Level 2
Radiated Electromagnetic Field	EC 801-3 Level1
Electrical Fast transient/Burst	EC 801-4 level 2
Safety Agency Approvals	UL Listed (UL 1950) CSA 22.2 #950 with D ₃ deviations IEC 950/EN 60 950 UL-94 flammability requirements for all computer boards

1.20 GPS Antenna System Specifications

Table 1-17 GPS Antenna Specification

Interfaces	
Timing Reference Interface: (Mechanical)	Deutsch 12 pin Receptacle – MMP24C-2212P1 or Equivalent
Electrical Characteristics:	RS422 Interface – 1 Input (Not Used), 2 Outputs: 1PPS, TSIP Serial Data and DC Power Input.
Cables	
Data / Power Cable:	Cable type - 4 TW PR 24AWG, Foil w/Drain, Plenum Rated FEP Teflon Jacket. Cable drawing #: (Honeywell) 51305554-x00 (Where x = length 1 = 50 ft, 2 = 100 ft, and 3 = 200 ft.)
Power	
Input Voltage:	9 to 32 VDC
Power Consumption:	250 mA @ 12VDC, 3.0 Watts Nominal.
Physical	
Dimensions:	5.8" W x 3.9" H (147 mm x 100 mm)
Weight:	13.4 oz. (0.38 kg)
Mounting:	Threaded Socket accepts 1.00" – 14 Straight Thread
Connector:	12 pin round, waterproof (See Interface section)
Performance	
General:	L1 frequency, C/A code (SPS), 8-channel, continuous tracking receiver.
Update Rate:	1 Hz
Accuracy:	Position: 25 m SEP without S/A Velocity: 0.1 m/sec without S/A
Timing Pulse:	Accuracy: UTC +/- 60 ns (without S/A) UTC +/- 100 ns (static) UTC +/- 300 ns (dynamic) Resolution: 40 ns
Acquisition: (typical)	Cold Start: < 2 minutes
Re-Acquisition:	< 1 second

1. Planning the TPN Bridge Platform
1.20. GPS Antenna System Specifications

Table 1-17 GPS Antenna Specification

Dynamics:	Velocity: 500 m/sec maximum
	Acceleration: 4g (39.2 m/sec ²)
	Jerk: 20 m/sec ³
Jam Immunity:	Resistant to jamming by a CW signal with power up to –86 dBm measured at the antenna element. The gain threshold increases to –56 dBm for jamming signals which are +/- 5 Mhz from the GPS carrier frequency.
Burnout Protection:	Protected from damage by RF signals at frequencies greater than 100 Mhz from the L1 frequency (1575.42 Mhz), when the power received by the antenna is no greater than one watt.
Environmental	
Operating Temperature	-30° to 70° C *
Storage Temperature	-55° to +100° C *
Operating Humidity	95% R.H. non-condensing
Storage Humidity	N/A
Operating Altitude	-400 m to +18,000 m *
Storage Altitude	N/A
Vibration	
Operating	0.25 G at 5 – 150 Hz all 3 axes, 60 minutes each axes. Reference spec. 51108010
Non operating	0.5 G at 3 to 200 Hz Reference spec 51108010
Shipping (Supplier carton)	10 to 500 Hz random, all 3 axes, 60 minutes each axes. Use “vertical” profile for each face of the carton; Reference spec. 51108010 Rev C section 5
Shock	
Operating	5G @ 18 ms (No failures in operation)
Contaminants	
Operating	G1 certified per ISA S71.04-1984

Table 1-18 GPS System Power Supply Specification

Input Range Voltage	85 to 264 Volts AC RMS
Line Frequency	47 to 63 Hz
Rated Input Voltage	100-120/200-240 Volts AC RMS
Inrush Current (max.)	30 A / 60 A peak @ 100 / 200 VAC
Input Current at rated input voltage and nominal output current.	0.24 A / 0.12 A @ 100 / 200 VAC
Output Voltages (nom) and Currents (rated)	+24V at 0.42 A
Output current min./max.	0...100 %
Efficiency @ rated input voltage & nominal output current.	75 %
Ripple-noise (Max.)	120 mVpp
Hold up time	15 ms at Rated Input V and Output Current
Over Current Protection	105 % of Rated Output. Automatic recovery required.
Rise Time (max.)	200 mS
Load regulation 0...100%	+/- 3 %

1.21 Backup Power

Attention

The TPN Bridge nodes perform a critical system function. It is the customer's responsibility to ensure that the reliability of the AC power to the TPN Bridge nodes is consistent with this important functionality. The primary and secondary bridge nodes should be on separate AC power systems, if possible, so that loss of AC power to one bridge node does not affect its redundant partner. Reference the *TPS System Site Planning* SW02-550 for further information on UPS.

1. Planning the TPN Bridge Platform

1.21. Backup Power

2. Installing the TPN Bridge

2.1 Overview

Scope

The installation process involves installing TPN Bridge nodes using one of two methods. The switch units are also mounted in the TPN cabinet.

Installation Method

Hot Insertion—For pre-existing system installation during continuing operation.

Cold Start—For pre-existing or new system installation that are not in operation and can be powered down.

Process description

Attention

In all cases, only break the two cables, Cable A and Cable B, at one place at a time. The Hot Insert procedure using the Hot Insertion Switchbox tool is to be performed only by trained Honeywell personnel.

Hot Insertion

⇒ Please refer to publication *TPN Bridge Overview and Implementation Guide* for the complete Hot Insertion procedure. It is contained in section "*Implementing the TPN Bridge System*."

Cold Start

- Place the TPN Bridge nodes and switch units in the customer's workplace according to the project plan.
- Ensure that all non-TPN Bridge nodes are powered down.
- Re-structure the LCN coax cable.
- Connect the TPN-Bridge nodes to the switch units.
- Configure the TPN Bridge software.
- Connect the LCN Cable A and B, and Tee's to the MAU for each Primary and Secondary TPN Bridge nodes.
- Verify proper operation of all nodes.

2. Installing the TPN Bridge

2.2. Installing TPN Bridges

2.2 Installing TPN Bridges

Scope

A TPN Bridge unit consists of:

- 2 Computer units
- 2 Ethernet switch units
- Shared peripherals (monitor, keyboard, mouse and peripheral switch)
- Ethernet cables
- Optional GPS Time Link system

Locate Equipment

Place the hardware and connect the peripherals to the computer as shown in Figure 2-2
To connect the peripherals refer to Figure 1-12.

Grounding TPN Bridge Nodes

The computer is grounded by the third wire in the power cord and by the grounded connector connected to an unused serial port connector. The cabinet is grounded (safety ground) in the power panel or building steel in an insulated 5 mm² (#4 AWG) wire.

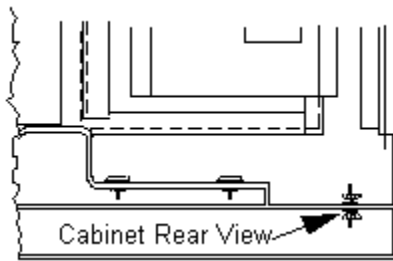


Figure 2-1 Ground Terminal in a TPN Cabinet

Connecting AC Power



WARNING

The power supply circuit is connected to ac power when the power cable is connected. The power control switch on the front panel only enables the power supply circuit outputs.

Configure the power select switch for the correct 120 or 240 Vac power before plugging in the ac power cord. Damage to the power supply will occur if this is incorrectly set.

2.3 TPN Bridge Cable Connections

GX200 Platform cable connections

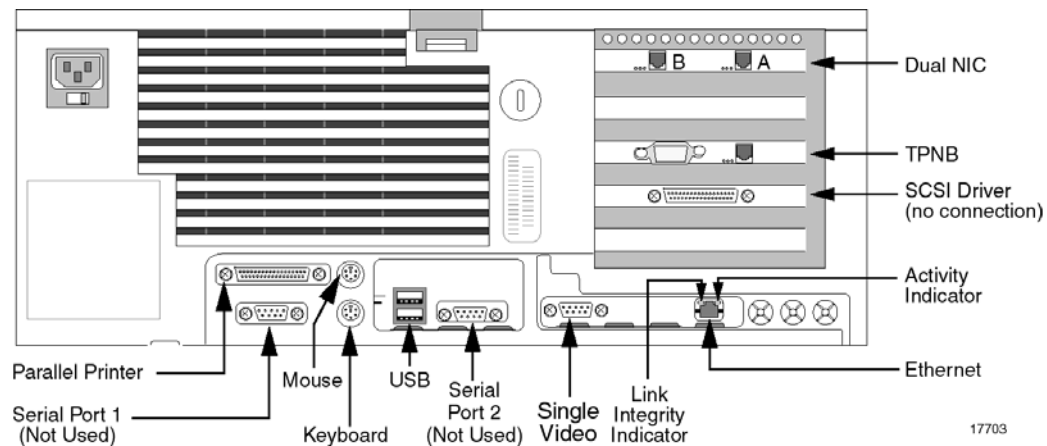
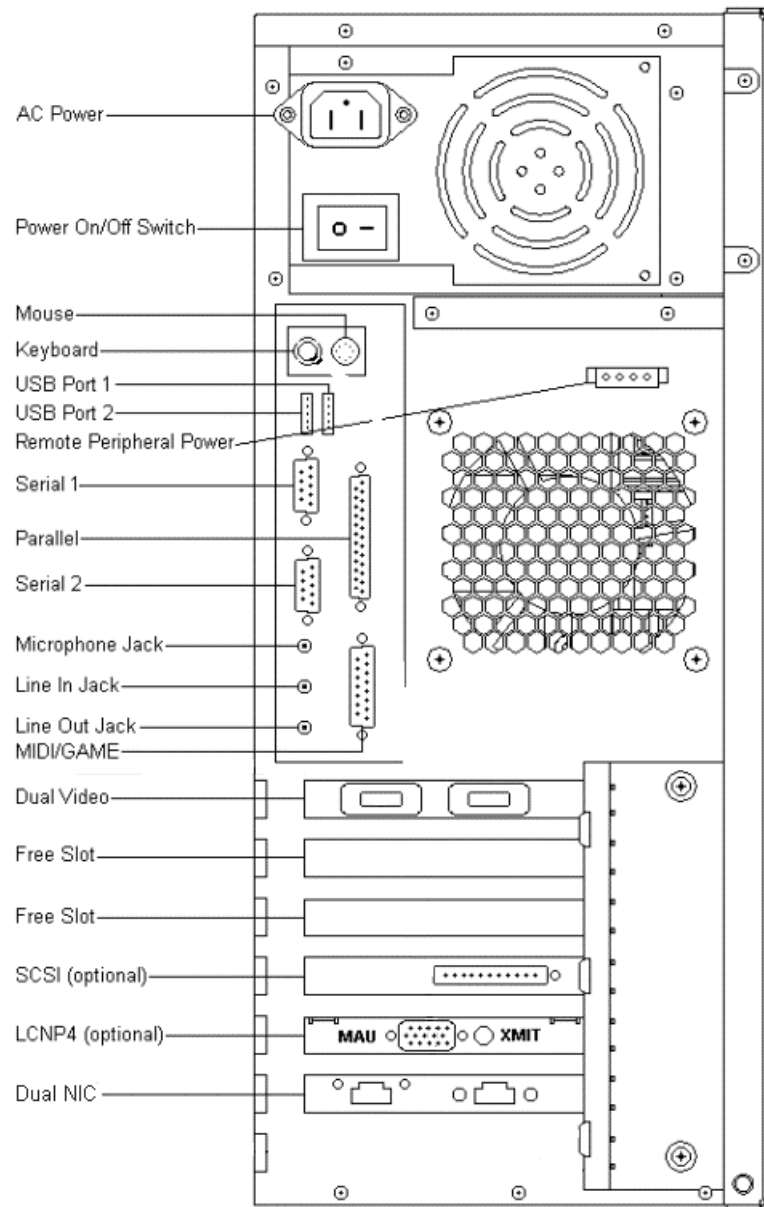


Figure 2-2 Cable Connections for GX200

2. Installing the TPN Bridge

2.3. TPN Bridge Cable Connections

WKS1 Platform cable connections



2. Installing the TPN Bridge

2.3. TPN Bridge Cable Connections

Figure 2-3 Cable Connections for WKS1

Dual NIC and LAN Cable connections

When connecting the LAN cables to the dual NIC adapter, verify the correct port is used. Figure 2-4 shows the differences in the two types of dual NIC adapters used for the TPN Bridge.

- Connect the Ethernet LAN Cable A (yellow boot) to the A port of the Dual NIC adapter.
- Connect the Ethernet LAN Cable B (green boot) to the B port of the Dual NIC adapter.

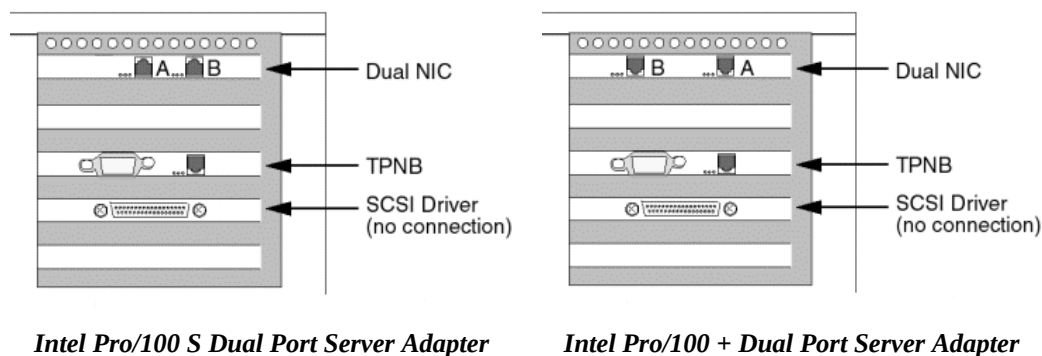


Figure 2-4 Dual NIC Adapters for TPN Bridge

2.4 AC Power

TPN Bridge computer



WARNING

The power supply circuit is connected to ac power when power is connected. The power control switch on the front panel only enables the power supply circuit outputs.

Configure the power select switch for the correct 120 or 240 Vac power before plugging in the ac power cord. Damage to the power supply will occur if this is incorrectly set.

TPN-Cabinet

The ac distribution bar has an ac breaker located at the rear, bottom of the cabinet. There is an optional ac distribution bar for a second power source (redundant power)

Attention

It is strongly recommended that the ac power cord be connected to a clean power source with backup such as an Uninterruptible Power Source (UPS). Refer to *TPS System Site Planning* and *TPN System Installation* manuals.



WARNING

The power supply circuit could be connected to ac power when power is connected to the power entry box. The power control switch on the front panel only enables the power supply circuit outputs.

Configure the power select switch for the correct 120 or 240 Vac power before plugging in the ac power cord. Damage to the power supply will occur if this is incorrectly set.

2. Installing the TPN Bridge

2.4. AC Power

2.5 Preparation for Installing the Ethernet Cable Network

Scope

The cables that connect each TPN Bridge node to the Bay Network 12-Port Ethernet Switch **should be identified** according to the source Bridge unit, terminating at the Switch port. For example, the left most cable shown below identified as C1ANP, translates as characters “C1” for Cluster (Segment) 1, “A” for Cable A, and “NP” for Primary Node. This labeling scheme is for illustration only. Customers may have their own unique labeling scheme.

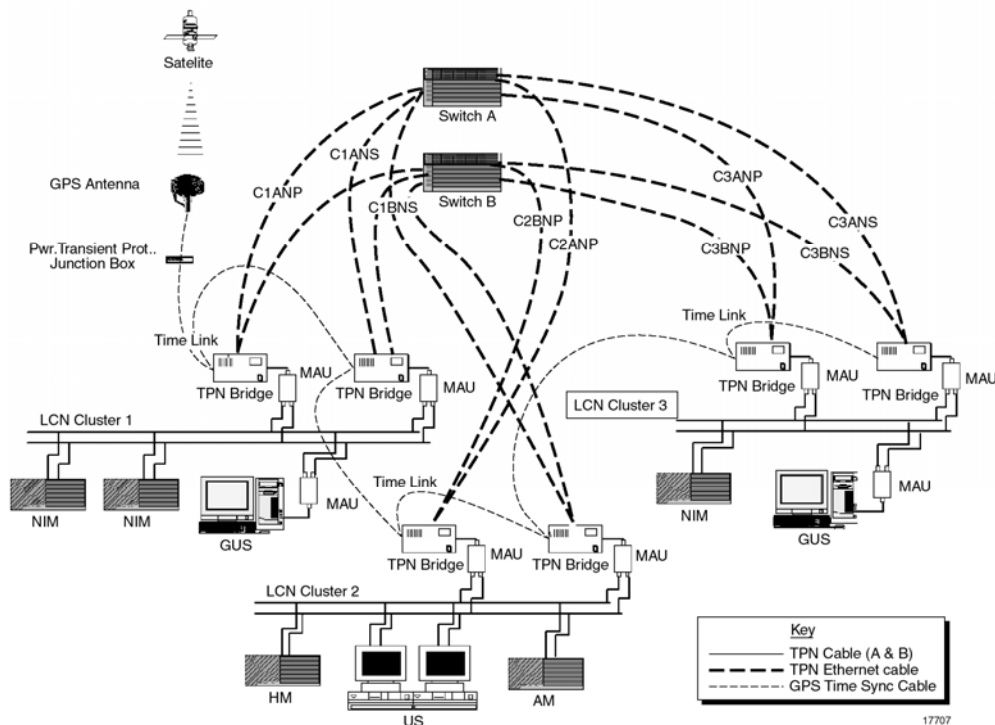


Figure 2-5 TPN Bridge Ethernet Cable Identification

2.6 Cable Connection Matrix

Scope

A cable connection template similar to the one in following table should be filled out and retained for customer installation.

Table 2-1 Cable Connection Matrix Template for TPN Bridge

Description		Switch A						Switch B					
SEGMENT #	Bridge / NIC Port	Port						Port					
		1	2	3	4	5	6	1	2	3	4	5	6
SEGMENT 1	Primary / A												
	Primary / B												
	Secondary / A												
	Secondary / B												
SEGMENT 2	Primary / A												
	Primary / B												
	Secondary / A												
	Secondary / B												
SEGMENT 3	Primary / A												
	Primary / B												
	Secondary / A												
	Secondary / B												

2.7 Installing the TPN Bridge

Procedure

Attention

The following procedure installs the monitor, keyboard, and mouse directly to the computer.

Refer to Figure 2-5 for cable connection locations.

Table 2-2 Connecting the Cables to the Computer

Step	Action
1	Refer to Table 2-1. Install the first cable listed there as stated in the Table.
2	Install the remainder of the cables listed in Table 2-1.
3	Connect the 104-keyboard cable to the keyboard connector on the rear panel of the computer as shown in Figure 2-2 or Figure 2-3.
4	Connect the mouse cable to the mouse connector on the rear of the computer shown in Figure 2-2 or Figure 2-3.
5	If a printer is to be connected, connect it to the printer connector on the rear panel of the computer shown in Figure 2-2 or Figure 2-3.
6	<u>Configuration</u> Configure the monitor for VGA sync using the switch on the rear of the monitor. Connect the monitor cable to the Video connector on the rear panel as shown in Figure 2-2 or Figure 2-3.

2. Installing the TPN Bridge

2.8. Installing the TPN Bridge in a TPN Cabinet

2.8 Installing the TPN Bridge in a TPN Cabinet

Scope

TPN Bridge computers can be installed in a TPN Cabinet using the TP-MOUNTR mounting kit. A 17-inch monitor, keyboard, and mouse assembly can be mounted in the same cabinet with a Shared Interface Mounting Kit, TP-HICAB1.

When several (2 to 4 computers are to share a monitor, keyboard, and mouse with multiple computers, a two-port switch, 51196084-100, or a four-port switch, 51196084-200, is used. See Figure 1-12.

Refer to Figure 2-5 for the topology information.

Use the installation parts from Table 2-3 to install the enclosure. The item numbers in the table are referred to in the procedure.

Table 2-3 Installation Parts (51403280) for the Computer Enclosure

Item	Part Number	Description	Qty
1	51403263-100	TPS Enclosure Assembly	1
2	51195168-616	Thread Forming Screw	4
3	51109933-003	Serrated Washer	4
4	51305468-100	Enclosure Mounting Support	1
5	51305468-200	Enclosure Mounting Support	1
6	51108393-006	Washer, Lock, Ext. Tooth	8
7	51108888-612	Black Machine Screw, Hex Head	4
8	51108389-004	Nut, Hex Lock, Nylon Insert	4
9	51305447-100	Wedge Handle Assembly	

Table 2-4 Installing the Computer Mounting Enclosure in a TPN Cabinet

Step	Action
1	Mount the Enclosure Mounting Support (Item 4), 51305468-100, mto the left side (from the front) to the left side vertical support with one of items 6 and 7.
2	Mount the Enclosure Mounting Support (Item 5), 51305468-200, mto the right side (from the front) to the right side vertical support with one of items 6 and 7.
3	Install the enclosure (item 1), 51403263-100, in the cabinet using 4 screws (Item 2) and washers (item 3).
4	Install two screws (item 7) on each side of the enclosure to fasten the enclosure to the mounting supports installed in steps 1 and 2.
5	Install two screws (item 2) and 2 washers (Item 3) on each side of the front of the enclosure to fasten the front of the enclosure to the back
6	Place the computer in the enclosure with the port connectors to the rear of the enclosure

2. Installing the TPN Bridge

2.9. Installing the GPS Antenna and/or Time Link Components

2.9 Installing the GPS Antenna and/or Time Link Components

Scope

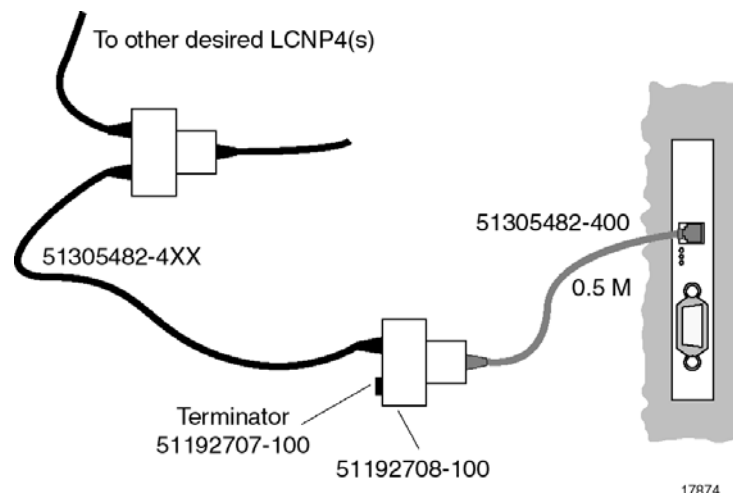
Installation includes the following:

- Time Link cables, cable Tee connectors, and cable modular terminator
- GPS Antenna, Antenna Mast, and cable
- Power/Surge Protection Junction Box

Install Time Link Cable Components

Table 2-5 Installing the Time Link Cable Components

Step	Action
1	Install a 0.5-meter Time Link cable from the TEE connector to the link port on the LCNP4 board.
2	Install the terminators.
3	Install the Time Link cable.
4	Repeat steps 1 through 3 until all desired LCNP4s have been connected. The last position gets an additional terminator or connects to the power/surge Junction Box if GPS option is utilized.



Install Power/Junction box (GPS Option)**Table 2-6 Installing the Power/Surge Protect Junction Box**

Step	Action
5	Mount the Power Surge/Junction Box to a point within 6 feet of building steel.
6	Connect a 6 foot long AWG #6 copper wire to the ground terminal (shown in Figure Figure 1-6) on the Power/Junction Box and to Building Steel. To safely and suitably conduct electrostatic transients back to earth, an AWG #6 copper wire is required.
7	Connect the ac power cord, 51305567-100, to the ac power connector on the box.

2. Installing the TPN Bridge

2.9. Installing the GPS Antenna and/or Time Link Components

Install GPS Antenna

Table 2-7 Installing the GPS Antenna

Step	Action
1	Locate the antenna mounting point that has a clear view of the sky, 30° above the horizon as shown in Figure 1-6.
2	Install the mast mounting plate to the surface (of the roof).
3	Mount the mast to the mounting plate
4	Install guy lines to stabilize the mast.
5	Mount the antenna onto the mast.
6	Install the antenna cable, 51305554-xxx, where xxx = tab 100 for 50 ft cable, tab 200 for 100 ft cable, tab 300 for 200 ft cable) Connect the cable to the antenna as shown in Figure 1-5.
7	Connect the other end of the antenna cable to J2 on the Power surge/junction Box.

2.10 Installing the Human Interface Switch

Scope

Installation will require either a 2-port or 4-port peripheral switch box. You will also need the appropriate connecting cables to interface the TPN Bridge computers.

The following models have been defined for installing the Human Interface (HI) Switch:

TP-RKVM4A KVM Switch, 4-Node, Cabinet/Rack Mounting

TP-DKVM2A KVM Switch, 2-Node, Desktop

TP-DKVM4A KVM Switch, 4-Node, Desktop

Each model number calls out the HI switch, the appropriate cable set, and also provides the mechanical parts to mount the HI switch, CRT, keyboard and mouse into its appropriate configuration.

2. Installing the TPN Bridge

2.10. Installing the Human Interface Switch

Connecting the User Console to the TPN Bridge

Table 2-8 Installing the HI Switch(es)

Step	Action
1	The CRT (monitor), keyboard and mouse connect to the HI switch box instead of the computer. Plug the keyboard cable into the port on the back of the unit in the "user console" section. Plug the mouse cable into the mouse port and the video cable into the video port.
2	Power-down all computers that will be part of your HI switch system.
3	Locate the appropriate cable kit to attach your first computer to the HI switch. Plug the keyboard cable from the kit into the connector of any available port on the HI switch. Ports are labeled A-B on the 2-Port and A-D on the 4-Port.
4	Plug the mouse cable into the appropriate connector of the same port. Depending upon the cable kit, you will have either a PS/2 style 6-pin mini DIN or a serial 9-pin D-shaped connector. Attach only one mouse per computer
5	Plug the kit's video cable into the monitor input connector. Plug the remaining end of each cable into the matching ports on your computer. Tighten all thumbscrews.
6	Repeat steps 3 through 5 for each additional computer to be attached and then power-up all attached computers.

Attention

The computers may be powered-up one-at-a-time or all at once. The green LEDs over each channel will light, indicating that the attached computer is powered on. After power-up, a lit amber LED indicates the selected computer. Other computers may now be selected for operation.

Selecting a Computer

There are two ways to select a computer. One way is via the Select push-button. This selects the next computer in sequence. Another way is by entering a short sequence of keystrokes on the keyboard. This is called keyboard, or hot-key, switching.

Table 2-9 Selecting a Computer by Switch Push-Button or Hot-Key

Step	Action
1	Press the Control key twice within one second to place your HI switch in Command Mode.
2	Verify that the keyboard LEDs are flashing indicating that you operating in Command Mode. In this mode, whatever is typed will be interpreted as HI switch system commands until Enter is pressed to accept the command or Escape is pressed to cancel.
3	In Command Mode, use Table below to type the letter(s) for the computer you wish to select. Press Enter to accept the new address (<i>Addr</i>).

Table 2-10 Hot-Key Sequence for Selecting a Computer

Key Sequence	Action
<CTRL><CTRL>Addr<Enter>	Selects an active channel via keyboard.
EXAMPLE	
1. <CTRL><CTRL>C<Enter>	Selects Channel C.
2. <CTRL><CTRL>B<Enter>	Selects Channel B.
3. <CTRL><CTRL>AB<Enter>	Selects the HI switch attached to Channel A on the base unit, then selects Channel B on the cascaded unit. (Not used on TPN Bridge)

3. Service

3.1 Service Policy

Definition of support

Attention

Dell provides supporting documentation for the standard items in the computer electronics unit. Honeywell provides supporting documentation for the items added to the standard unit by Honeywell.

Electronic chassis

Service support has been defined as follows:

Replacing the failed item with one of the following from Dell supports any failure in the electronics chassis:

Item	Item	Item
Mother Board	RAM Memory	SCSI Board
Floppy Drive	CD-ROM	Hard Drive(s)
Riser Assembly	Computer Keyboard	Mouse
Power Supply		

The exception to this is for the optional items provided by Honeywell:

Item	Item	Item
17-Inch Desktop Monitor	Tape Drive (optional)	Zip Drive (optional)
Dual NIC board	TPNB board	

3.2 Board Placement

Scope

The PCI and ISA boards are located as follows:

Table 3-1 Board Placement

Slot Number	Board Identification
ISA 1	
PCI 5	Dual NIC
PCI 4	
PCI 3	TPNB
PCI 2	SCSI Controller
PCI 1	

Board Placement

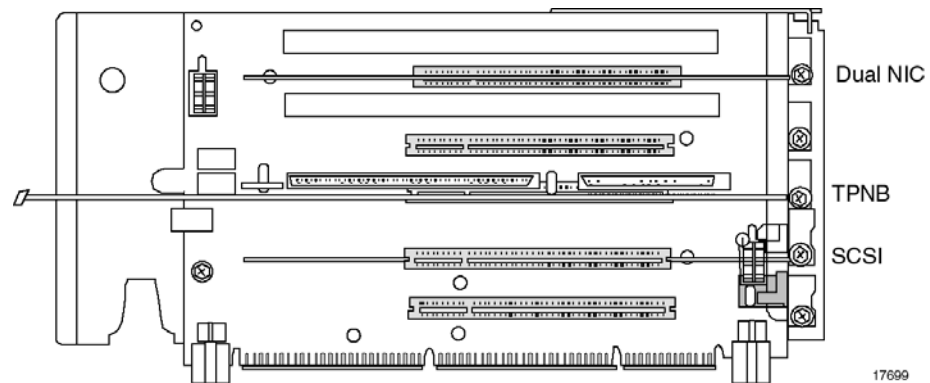


Figure 3-1 Board Placement in TPN Bridge

3.3 Troubleshooting

Scope

Service for the Dell computer is provided by the Dell Systems Service publication.

The Honeywell-provided equipment is supported by this publication.

Verifying Correct System Interrupt Assignments

To verify the correct system interrupt assignments, complete the following:

Table 3-2 Verifying System Interrupt Assignments

Step	Action
1	From the NT 4.0 START menu, choose <u>Programs</u> > <u>Administrative Tools</u> <u>Common</u> > <u>Windows NT Diagnostics</u> .
2	From the dialog box that appears, choose the Resources tab.
3	Ensure that the IRQ button is selected at the bottom of the window, and verify the proper system interrupt assignments from Table 3-3.
4	If the listed IRQ assignments do not agree with Table 3-3, it may be necessary to force proper assignments through the Plug and play BIOS. This may be necessary in the case of a motherboard replacement.

3. Service

3.4. Removing and Replacing the Computer in the Cabinet

Table 3-3 System Interrupt Assignments

IRQ	Device	Notes
01	MSI 8042	Computer Keyboard Port
04	Serial	Serial port 1 on motherboard COM 1
05	Serial	COM 3 on IKB12
06	Floppy	Floppy
09	cs32bal11	On board Crystal sound
*	aic78xx	Adaptec 2940UW2 SCSI controller
*	DC21X4	Ethernet with AUI
*	tdcaint	LCNP
12	MSI 8042	Mouse Port

These PCI Bus devices are dynamically reconfigurable by the system. The actual IRQs assigned may change from one boot to the next. They may even show overlapping IRQ assignments among themselves. This is normal.

3.4 Removing and Replacing the Computer in the Cabinet

Foam insulators

There are foam insulators and gaskets used between the computer and the cabinet mounting chassis. These insulators provide protection against vibration and ensure proper insulation for the computer. Making sure these insulators are positioned properly and are not damaged when removing or replacing the computer in the mounting chassis is critical.

Remove the Computer from the Cabinet



CAUTION

When removing the TPN Bridge computer in the cabinet, **DO NOT** slide the computer from the cabinet's mounting chassis. Doing so can destroy the foam insulators and cause a complete ISOLATION.

Table 3-4 Removing the Computer from the Cabinet

Step	Action
1	Shut down the Windows NT operating system.
2	Turn off the ac breaker for the cabinet and disconnect all the ac power cords from the breaker box.
3	Disconnect the LCN coax cable T-connectors from the MAU assembly.
4	Remove the front and rear covers for the mounting chassis.
5	Disconnect the cables from the rear of the computer.
6	If the mounting chassis for the computer does not have a node or chassis mounted directly above it, go to Step 9. Otherwise, continue with Step 7.
7	Remove all the screws that are securing the mounting chassis to the cabinet.
8	Remove the mounting chassis (with the computer) from the cabinet.
9	Remove the screws from the top cover of the mounting chassis and lift the cover off.
10	Remove the piece of isolation foam from the top of the computer.
11	Lift the computer out of the mounting chassis.

3. Service

3.5. Fan Replacement

Replace the Computer in the Cabinet



CAUTION

When replacing the TPN Bridge computer in the cabinet, **DO NOT** slide the computer into the cabinet's mounting chassis. Doing so can destroy the foam insulators and cause a complete ISOLATION.

Table 3-5 Removing the Computer from the Cabinet

Step	Action
1	Place the computer in the mounting chassis.
2	Place the piece of isolation foam on top of the computer.
3	Place the top cover of the mounting chassis and secure it with the screws.
4	If you did not remove the entire mounting chassis from the cabinet, go to Step 7. Otherwise, continue with Step 5.
5	Replace the complete mounting chassis (with the computer) back in the cabinet.
6	Secure the mounting chassis to the cabinet with the screws.
7	Reconnect the cables to the rear of the computer.
8	Replace the front and rear covers for the mounting chassis.
9	Reconnect all cables.

3.5 Fan Replacement

Procedure

The fan is located on the rear panel of the chassis beside the power supply. To remove it see the “Remove the Chassis Fan” found in the Dell System Service publication.

4. Troubleshooting, Cleaning, and Pinning Drives

4.1 Troubleshooting an Inoperative Drive

Failure modes

In general, if only one drive fails, the drive is most likely at fault. If both drives are inoperative, check the power supply in the electronics module and the power extension cable from the computer to the drive enclosure.

Isolation procedure

Table 4-1 Troubleshooting a Drive Problem

Step	Action
	If both drives fail, disconnect the power cable coming out of the rear of the electronics module; +5 Vdc and +12 Vdc must be present.
	Check for power at the input to the drive enclosure and at the input to the drive.
	If no problem is found in Step 1, check that all cable connections are seated.
	Replace the drive with a spare.
	If all of the above failed to locate the problem, check the SCSI cable inside the electronics module.
	The last resort is to replace the motherboard inside the electronics module.

Illustration

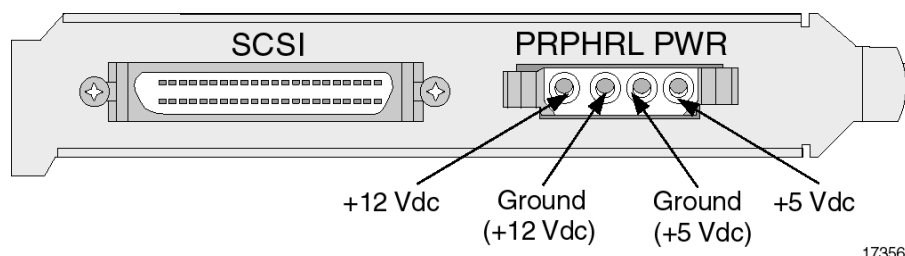


Figure 4-1 Remote Peripheral Connector Pin-out

4. Troubleshooting, Cleaning, and Pinning Drives

4.2. Remove a Zip Cartridge with Power Off

4.2 Remove a Zip Cartridge with Power Off

Scope

The Zip cartridge can be removed only when power is off by accessing the rear of the drive and firmly inserting a small object like a straightened paper clip into the small hole on the rear of the drive, then extracting the cartridge.

4.3 Pinning the Drive SCSI Address

Scope

CD-ROM, tape, and Zip drives have unique SCSI addresses. Each drive has a label that details the method of pinning all eight possible addresses. The unique address for each drive is shown in Table 4-2.

There can be a primary and secondary ZIP drive or a primary and secondary tape drive in a tower computer where there are three drives possible.

Table 4-2 SCSI Drive Addresses

Drive	Primary Drive Address	Secondary Drive Address
CD-ROM	5	5
Tape	4	6
Zip	6	4

5. Servicing Peripheral Drives

5.1 Overview

Scope

The CD-ROM, tape, and Zip drives have a SCSI interface and are located in the peripheral slots on the front of the computer. The SCSI addresses for the drives are given in Table 4-2.

5.2 Pinning the Drive SCSI Address

Drive SCSI address pinning

The drives have a SCSI Bus address shown in Table 4-2.

5. Servicing Peripheral Drives

5.3. Replacing the Drive

5.3 Replacing the Drive

Procedure

Table 5-1 Replacing a Drive

Step	Action
1	Ensure that the media has been removed from the drive.
2	Perform a system shutdown. Power off the unit.  53895 Remove the power cord.
3	Remove the chassis cover by pressing the button on each side and rotating the cover off the chassis.
4	Disconnect the dc power cable from the back of the drive.
5	Use ESD  Wrist Strap 16310 Disconnect the data cable from the back of the drive.
6	Squeeze the tab, located on each side of the front of the tape drive, toward the center and pull the drive assembly out of the chassis.
7	Remove the screws attaching the drive to the mounting plate.
8	Ensure the SCSI address defined in Table 4-2.is pinned on the back of the drive. Use the pinning method shown on the label on the drive.
9	Mount the new drive to the plate.
10	Slide the new drive assembly into the slot in the chassis until it clicks in place.
11	Reconnect the dc power cable and the data cable to the back of the drive.
12	Replace the chassis cover.
13	Reconnect ac power cord and power up the unit

6. Drive Service

6.1 Overview

Scope

Servicing of the hard drives and floppy drives is covered in the Dell documentation for your specific TPN Bridge computer.

References

Refer to the *Dell Optiplex GXa Service Manual* (51555) for information on servicing components in the Dell chassis.

6. Drive Service

6.1. Overview

7. 17-Inch Samsung Monitor Service

7.1 Overview

Scope

This section describes the operational controls of the Samsung 17-inch monitor (SyncMaster 17GLI) shown below.

7.2 Operational Controls

Front Controls

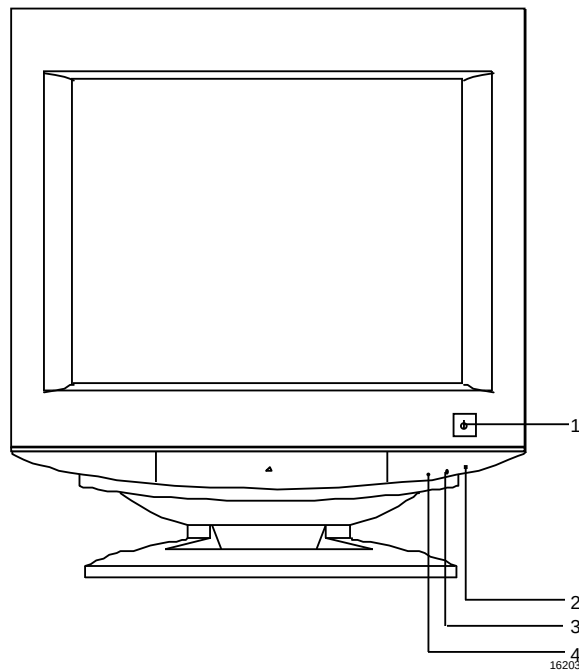


Figure 7-1 Samsung SyncMaster 17-Inch Monitor Front View

Front control panel

The numbers in the illustration below refer to the location designated in Table 7-1.

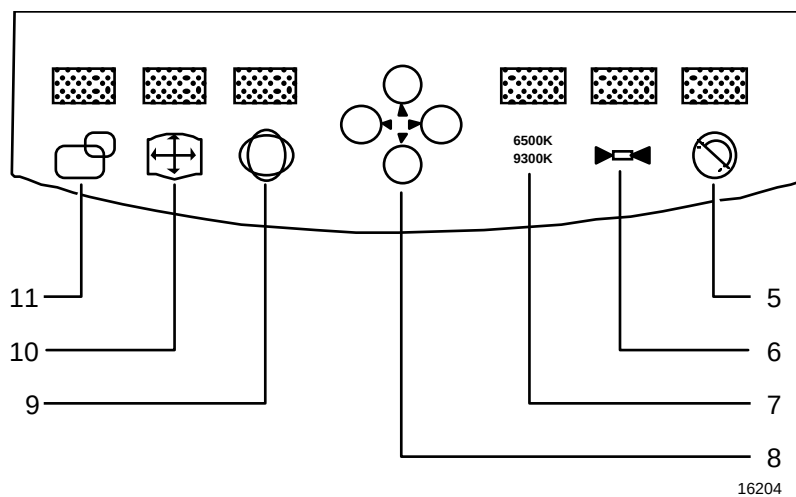


Figure 7-2 Front Panel Controls

Controls key table

The numbers in the Location column relate to the number in Table 7-1 and Figure 7-2.

Table 7-1 Key to Monitor 17-inch Controls

Location	Symbol	Description
1.		Power Button (Push)
2.		Power Indicator
3.		Contrast Control
4.		Brightness Control
5.		Degauss button
6.		Recall Button
7.		Color Temperature Control
8.		Adjustment Control
9.		G/D (Geometric Distortion)
10.		Size (H/V) Information
11.		Position (H/V) User, Preset Mode

7.3 Operational Controls Description

Scope

The following adjustments are basic operator controls and can be adjusted to suit the user.

Power button

The power button turns the monitor on and off. Push the power button once to turn the monitor on; the power LED lights when the power is on. Push the button again to turn the monitor off.

Power indicator

This dual color indicator shows the state of your monitor.

- When the monitor is operating normally, the LED indicator is green.
- When the monitor is in one of the power saving modes, the indicator is orange or blinks orange and green.

Contrast control

The contrast control adjusts the degree of difference between dark and light areas of the image.

Brightness control

The brightness control adjusts the overall brightness of the display image.

Adjustment control

Use the adjustment control buttons to adjust the display image while the control function is active (the indicator is orange); however, you cannot adjust the display image while a control function is disabled (indicator is green).

Push the up or right button to increase the value of the adjust function. Push the down or left button to decrease the value of the adjustment function.

Attention

The monitor automatically saves your preferred display setting 3 to 4 seconds after the last adjustment.

8. Ethernet Service

8.1 Troubleshooting

Scope

There are two indicators that are located in the RJ-45 connector on the back of the computer.

The left indicator is the Link Integrity Indicator and the right indicator is the Activity Indicator.

Illustration

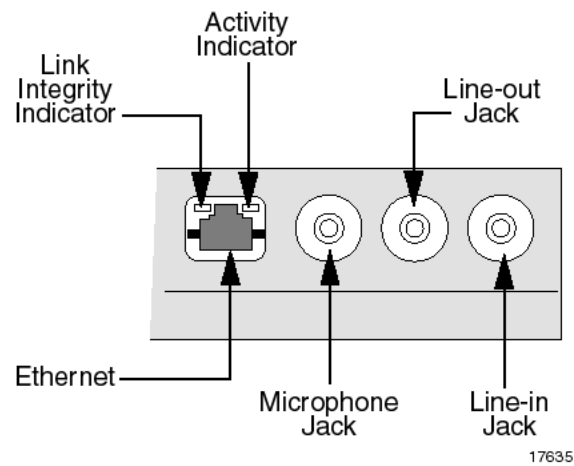


Figure 8-1 Ethernet Indicators on the RJ-45 Connector

8. Ethernet Service

8.1. Troubleshooting

9. Replacing and Servicing Parts

9.1 Power Supply

Overview

Refer to the Dell Optiplex GXa (51555) Service Manual for information on servicing components in the Dell chassis..

9.2 Dual NIC Adapter

Replacing the Dual NIC adapter

The dual NIC network adapter card originally installed in TPN Bridge nodes has been replaced with a different model. When replacing the TPN Bridge network card, you will need to update the driver for the network adapter card and configure the card's MAC addresses. Refer to "Replacing the Network Card" in Section 5 of the *TPN Bridge Overview and Implementation Guide*. See also the latest SCN.

Updating the Dual NIC adapter driver

Table 9-1 **Updating Dual NIC adapter driver**

Step	Action
1	Insert the TPN Bridge Software CD into the CD-ROM drive and cancel the automatic install routine.
2	Copy the entire Intel folder from the TPN Bridge Software CD to your local hard drive. ATTENTION: It is very important to copy the entire folder along with its sub-folders.
3	Select Start > Settings > Control Panel .
4	Double-click Network .
5	Select the Adapter tab from the Network dialog.
6	Select one of the network adapter ports for the dual NIC adapter card, and click Update .
7	From the Windows NT Setup dialog, type in the location you copied the Intel folder to followed by the complete driver location path, which is: <i>Intel\network\disk1</i> For example: <i>C:\Intel\network\disk1</i>

9. Replacing and Servicing Parts

9.2. Dual NIC Adapter

Table 9-1 Updating Dual NIC adapter driver

Step	Action
8	Click Continue and then close the Network dialog box.
9	Restart the computer when prompted.
10	To verify the driver was updated, perform the following: a) Select Start > Settings > Control Panel b) Double-click Network. c) Select one of the network adapter ports for the dual NIC adapter card, and click Properties. d) Select the Network Driver tab from the Intel ProSet dialog. e) Review the description field under Driver Information to verify the driver version.

9.3 Spare Parts

WKS1 Platform

These spare parts are for parts located in the computer chassis and the TPN cabinet.

Table 9-2 PC Electronics Spare Parts for WKS1

Description	Part Number
Tee Connector, LCN Coax	51190728-100
Label, Yellow LCNA	51195787-102
Label, Green LCNB	51195787-103
17-Inch Multi-synch CRT	51196793-200
21-Inch Desktop Monitor (Sony)	51197014-100
Intel Pro 100S Dual Port Network Adapter Note: this adapter replaces the Intel Pro 100+	51197085-200
Adaptec 29160N SCSI PCI Controller Card	51197182-906
1.44 MB Floppy Drive CDI	51199171-903
SXGA Dual Head 16MB AGP Video Card	51199171-905
Adaptec 2100S RAID controller	51199171-906
20 GB ATA-100. 7200 RPM Hard Drive or equivalent	51199171-907
18.1 GB SCSI 10K RPM Hard Drive or equivalent	51199171-908
10/100 Base Tx NIC	51199171-909
ATX Motherboard V2	51199171-910
256MB SDRAM DIMM ECC, PC-133	51199171-912
Fan Assembly	51199171-913
Air Filter	51199171-914
ATX Motherboard V5	51199171-918
36.7 GB SCSI 10K RPM Hard Drive or equivalent	51199171-919
40 GB ATA-100. 7200 RPM Hard Drive or equivalent	51199171-922
Power Supply, 200-250 Watt	51199171-923
CD-ROM, EIDE 40 x 12 x 48	51199171-924
1.266 GHz, FCPGA2, Pentium III processor kit	51199171-925
Drive Bay Filler Bracket	51199171-926

9. Replacing and Servicing Parts

9.3. Spare Parts

Table 9-2 PC Electronics Spare Parts for WKS1

Description	Part Number
Internal to External Power Cable	51199171-928
LCN MAU Assembly	51305348-100
LCN MAU Cable	51305380-100
Power Supply Adapter Plate	51305718-100
TPNB Assembly without Bracket	51403299-300
TPNB Assembly with Bracket	51403299-400

GX200 Platform

These spare parts are for parts located in the computer chassis and the TPN cabinet.

Table 9-3 PC Electronics Spare Parts for GX200

Description	Part Number
Battery (Panasonic CR2032)	
Tee Connector, LCN Coax	51190728-105
Label, Yellow LCNA	51195787-102
Label, Green LCNB	51195787-103
17-Inch Multi-synch CRT	51196793-200
Zip Drive	51196929-135
Matrox Productiva Dual Video Board	51196988-100
21-Inch Desktop Monitor (Sony)	51197014-100
Intel Pro 100S Dual Port Network Adapter Note: this adapter replaces the Intel Pro 100+	51197085-200
PC-Based System for Desktop/Console, Pentium III 866MHz	51198718-102
Microsoft Intellimouse	51198718-900
WIN95 Space Saver Quiet Key, 104 Key Keyboard	51198718-901
Power Supply, 200 W (GX200)	51198718-903
128 MB RDRAM, ECC, RIMM, PC700 compliant	51198718-904
1.44 MB Floppy Drive (GX200)	51198718-905
CD-ROM Drive, SCSI, 680M, 40X or equivalent	51198718-906
Adaptec 29160N SCSI PCI Controller	51198718-907
9.1 GB Ultra 160 SCSI Hard Drive, 10K RPM	51198718-908
Motherboard (OptiPlex GX200)	51198718-912
866 MHz Desktop Pentium II Processor Kit	51198718-952
LCN MAU Assembly	51305348-100
LCN MAU Cable	51305380-100
TPNB Assembly without Bracket	51403299-300
TPNB Assembly with Bracket	51403299-400

9. Replacing and Servicing Parts

9.3. Spare Parts

GX1 Platform

These spare parts are for parts located in the computer chassis and the TPN cabinet.

Table 9-4 Computer Electronics Spare Parts GX1

Description	Part Number
Battery (Panasonic CR2032)	
Tee Connector, LCN Coax	51190728-100
Power Cord, 220V, 2.5 Meter	51190921-210
Label, Yellow LCNA	51195787-102
Label, Green LCNB	51195787-103
17-Inch Multi-synch CRT	51196793-200
500 MHz Desktop Pentium II Processor Kit	51196982-102
Microsoft Intellimouse	51196982-900
Keyboard 104	51196982-901
Power Supply, 200 W (GX1)	51196982-903
128 MB SDRAM, ECC, DIMM, PC100 compliant	51196982-904
1.44 MB Floppy Drive (GX1)	51196982-905
600 MB CD-ROM Drive 17/40x or equivalent	51196982-906
Riser Assembly, 5 Slot, GX1 console mount	51196982-909
Filler Plate	51196982-910
500 MHz Motherboard (OptiPlex GX1)	51196982-912
2940U2W UltraWide SCSI Adaptec PCI Controller	51196982-913
9.1 GB SCSI Ultra Wide U2W Hard Drive, 1-Inch	51196982-914
21-Inch Desktop Monitor (Sony)	51197014-100
Intel Pro 100S Dual Port Network Adapter Note: this adapter replaces the Intel Pro 100+	51197085-200
LCN MAU Assembly	51305348-100
LCN MAU Cable	51305380-100
TPNB Assembly without Bracket	51403299-300
TPNB Assembly with Bracket	51403299-400

Ethernet Switch replacement unit

The unit is configured for Honeywell for this application.

Table 9-5 Unique Ethernet Switch Spare Parts

Description	Part Number
12 Port 10/100BaseT Ethernet Switch	51197083-100*

10. Glossary

10.1 Communications Terminology

Specialized terminology used in this document includes:

10 BaseT — 10 Mbit Ethernet using wire connected as a “Star” topology with Hubs or Switches

100 BaseTX — 100 MBit Ethernet using UTP or STP

Asynchronous Transfer Mode (ATM)—a communication protocol based on “cell” switching

CAT 3 Cable — widely deployed twisted pair cable designed for low and medium speed audio and digital communications

CAT 5 Cable — computer grade twisted pair cable specified for higher speed digital communications

COTS—*Commercial Off The Shelf*

Ethernet—*a high speed Local Area Network based on collision detection media access*

Fast Ethernet — 100 MBit version of Ethernet

Hub — the center element of an Ethernet “star that acts as a shared media cable

Network Interface Card (NIC) — a PCI or ISA board for connection to an open network

Shielded Twisted Pair (ST) —shielded computer cable

Switch —the center element of an Ethernet or ATM Star that acts like a telephone exchange

Unshielded Twisted Pair (UTP)— unshielded twisted pair cable

Electronics and Packaging Terms

Accelerated life Testing (ALT) — a technique to project product life from test results

CE-Mark —an indication of compliance to European Union Standards

Technical Compliance File (TCF)— a CE-Mark test to a configuration instead of standard

10.2 Computer System Terms

COTS — Commercial Off The Shelf

NT Server — WindowsNT Server operating system

NT Workstation — WindowsNT Workstation Client operating system

Workstation — a high performance computing platform

10.3 Honeywell Terminology

Console — Ergonomic furniture and packaging of control room equipment

Global User Station (GUS) — the TPS human interface workstation

Local Control Network (LCN) — Honeywell proprietary time-critical supervisory LAN

Packaged Plant Control Network — an open LAN system for industrial control

Total Plant Network (TPN) — a control room system

Total Plant Solution (TPS) — a critical control room supervisory and process connected system

11. Index

1

1.10 Equipment Configuration Rules, 39

17-Inch Monitor

Controls, 91

Operation, 91

C

Cable Connection Matrix, 69

Cabling

Interconnect Components, 70

Configurations for TPN Bridge, 47

Custom Cable Length, 32, 37

D

Description

Ethernet 100 BaseTX Cable, 31, 36

Drive troubleshooting, 85

Dual NIC Adapter, 97

Dual NIC and LAN Cable connections, 66

E

Equipment Specifications, 50

Ethernet Service, 95

Ethernet Switch Specifications, 55

F

Fan Replacement, 84

G

GPS Antenna and/or Time Link
Components installation, 73

GPS Antenna System Specifications, 57

GX200 Platform cable connections, 64

H

Human Interface Switch, 45

Human Interface Switch installation, 76

I

Installation

Grounding, 62

TPN Bridge, 70

M

Monitor

Samsung, 17-Inch, 91

P

Pinning the Drive SCSI Address, 86

R

Regulatory Compliance, 53

Removing and Replacing the Computer in the Cabinet, 82

Replacing

Computer in Cabinet, 82

S

Samsung Monitor, 91

Service

Board Location, 80

Definition of Support, 79

Support, 79

Troubleshooting, 81

Spare parts, GX1, 102

Spare parts, GX200, 101

Spare parts, WKS1, 99

Specifications

GX1 GX1, 50

GX200, 50

Minitower Computer, 50

Standard Cable Length, 32, 37

Switch Module Positioning, 48

T

TPN Bridge Configuration Rules, 43

Troubleshooting, 81

BIOS Interrupt Assignments, 81

Inoperative Drive, 85

W

WKS1 Platform cable connections, 65

Honeywell

Honeywell International
Process Solutions
2500 West Union Hills
Phoenix, AZ 85027